Aquaporin structural archetype and its modulations







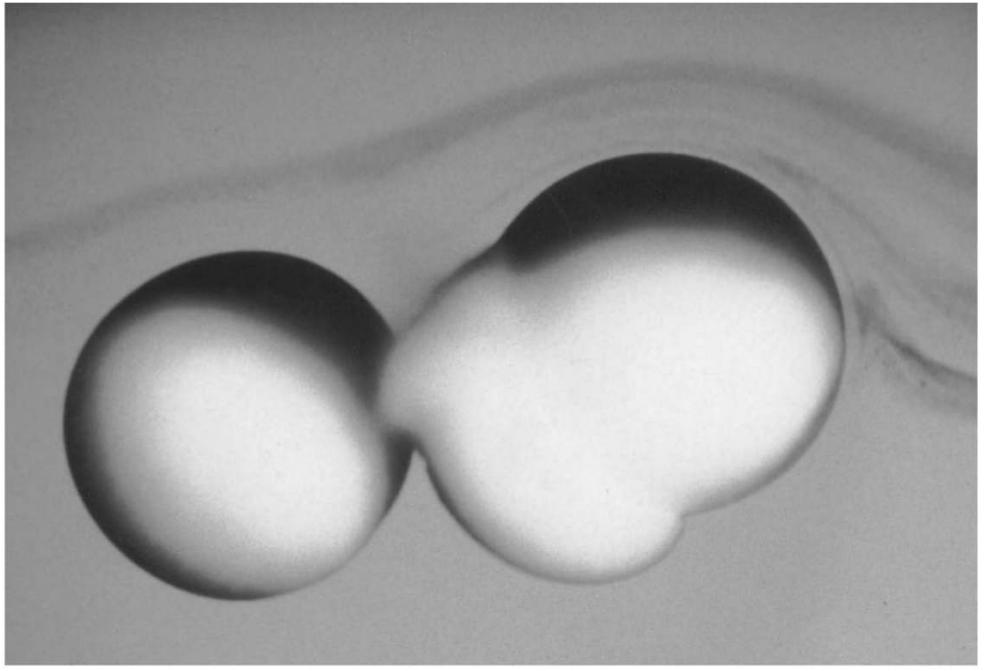
Daniel Auguin, Stéphane Réty, Vincent Mocquet

1.1. Aquaporins : water channels and more
1.2. AQP structural archetype: the core functional unit
1.3. AQP structural archetype: simple design but high functional diversity
1.4. Molecular dynamics simulations

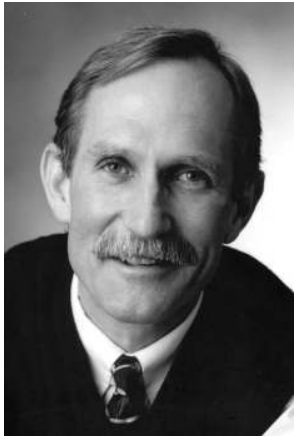
2.1. The plant case : a focus on intra-cellular gating
2.2. Modulation by steroids : the rôle of extra-cellular loops
2.3. Modulation by cations: highlight on cooperativity between monomers

3. Conclusions and perspectives

1.1. Aquaporins : water channels and more



Preston, G. M., T. P. Carroll, W. B. Guggino, and P. Agre. 1992. "**Appearance of Water Channels in Xenopus Oocytes Expressing Red Cell Chip28 Protein.**" Science (New York, N.Y.) 256 (5055): 385–87.

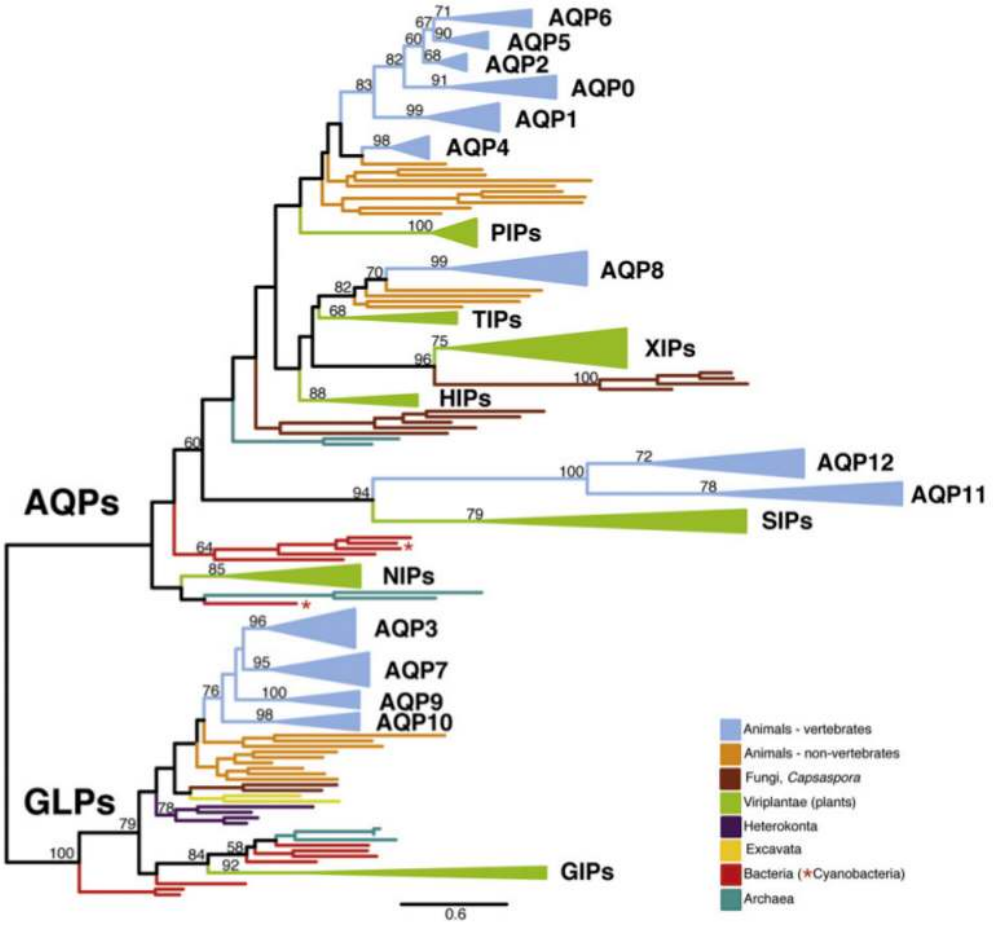


Peter Agre

The Nobel prize in Chemistry 2003

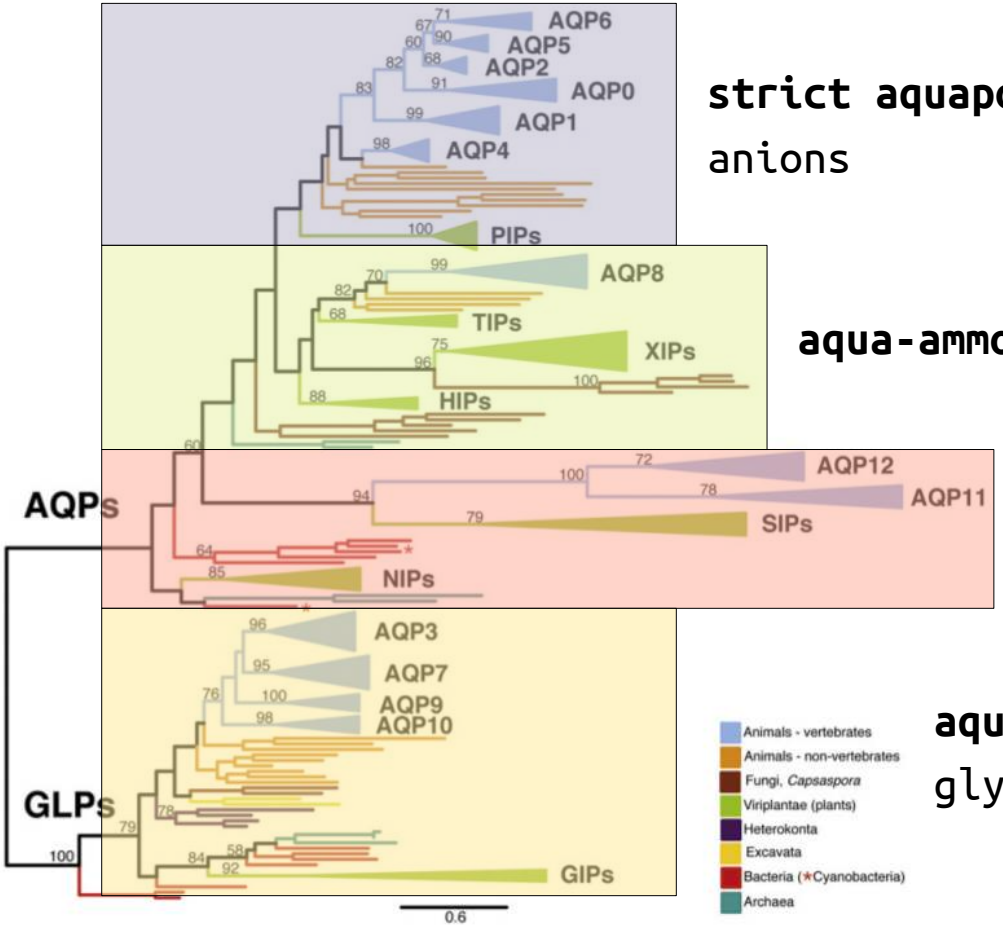
Prize motivation: "for the discovery of water channels"

1.1. Aquaporins : water channels and more



Abascal, F., Irisarri, I., Zardoya, R., 2014. **Diversity and evolution of membrane intrinsic proteins.** Aquaporins 1840, 1468–1481. https://doi.org/10.1016/j.bbagen.2013.12.001

1.1. Aquaporins : water channels and more



strict aquaporins (AQP) → H_2O, H_2O_2 and (inorganic) anions

aqua-ammoniaporins → H_2O, H_2O_2 and NH_3/NH_4^+

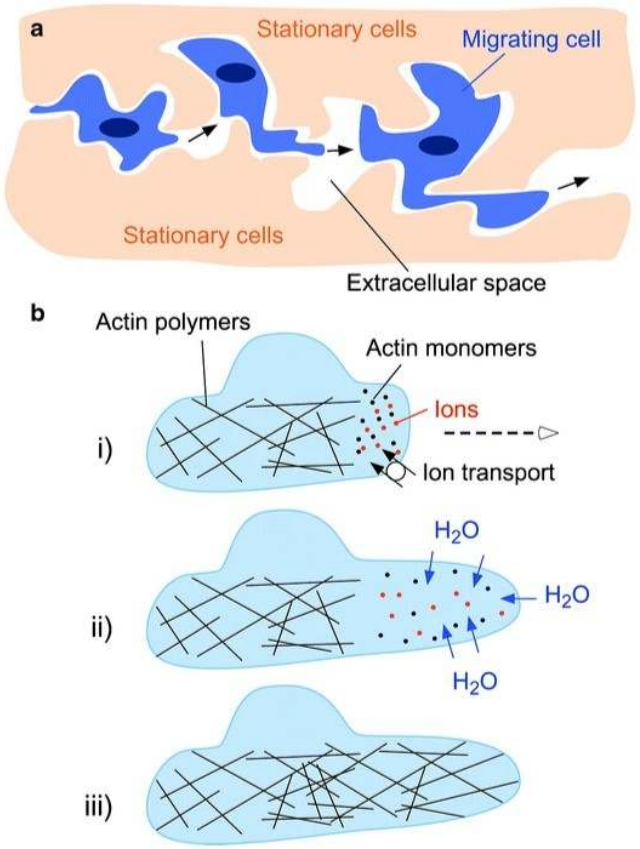
super-aquaporins → H_2O, H_2O_2 and (organic) anions

aqua-glyceroporins (AQGP) → H_2O, H_2O_2, glycerol and (organic) anions

Abascal, F., Irisarri, I., Zardoya, R., 2014. **Diversity and evolution of membrane intrinsic proteins.** Aquaporins 1840, 1468–1481. https://doi.org/10.1016/j.bbagen.2013.12.001

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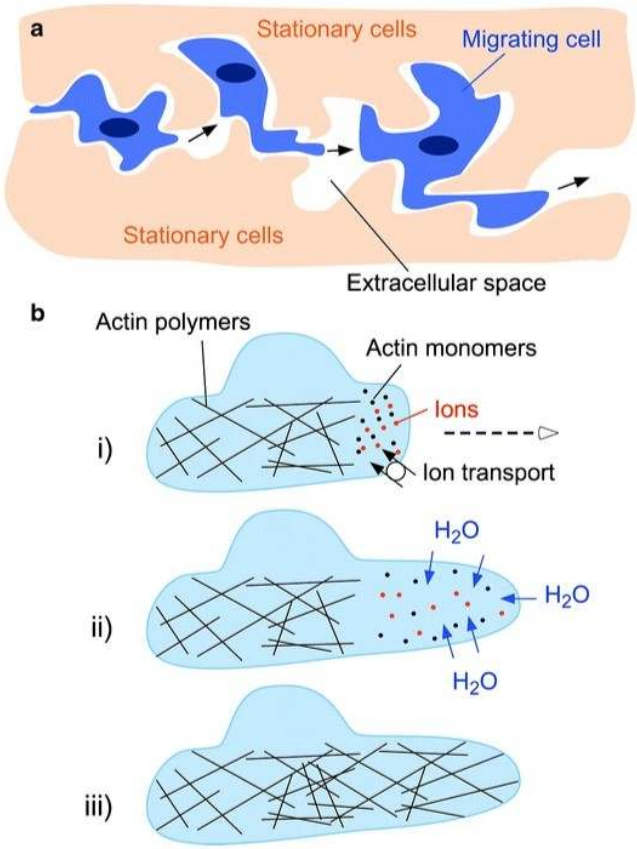
Cell homeostasis / volume changes / proliferation / motility



Papadopoulos, M.C.; Saadoun, S.; Verkman, A.S. **Aquaporins and Cell Migration.** *Pflugers Arch - Eur J Physiol* **2008**, *456*, 693–700, doi:10.1007/s00424-007-0357-5.

1.1. Aquaporins : water channels and more

Cell homeostasis / volume changes / proliferation / motility



→ Tumor cells express AQPs and a correlation exists between tumor grade and AQP expression.

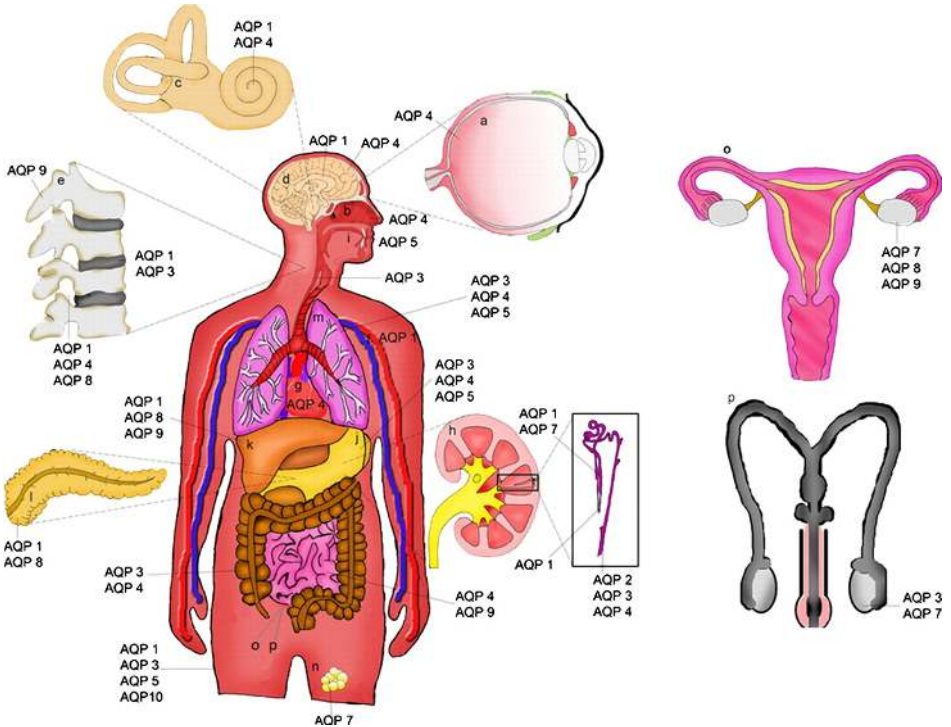
→ AQPs expression correlates with tumor cell migration and metastatic potential.

Ribatti, D.; Ranieri, G.; Annese, T.; Nico, B. **Aquaporins in Cancer.** *Biochimica et Biophysica Acta (BBA) - General Subjects* **2014**, *1840*, 1550–1553, doi:10.1016/j.bbagen.2013.09.025.

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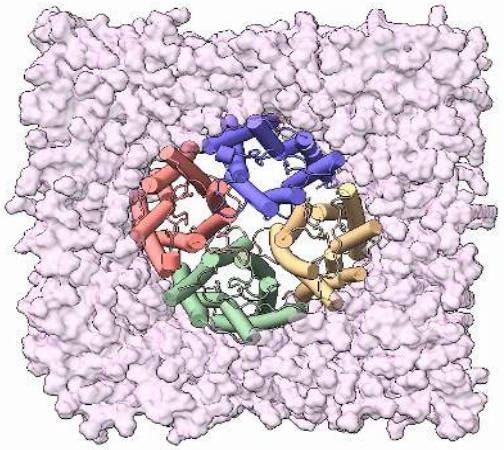
Cell homeostasis / volume changes / proliferation / motility

Trans-cellular water flows in organs and tissues

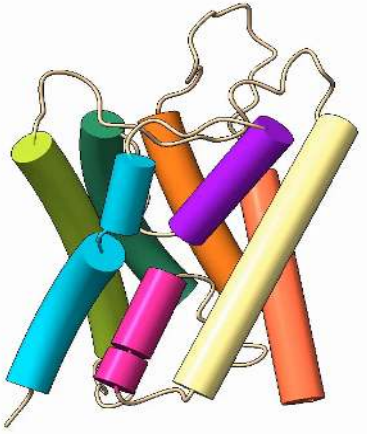


Day, R.E.; Kitchen, P.; Owen, D.S.; Bland, C.; Marshall, L.; Conner, A.C.; Bill, R.M.; Conner, M.T. **Human Aquaporins: Regulators of Transcellular Water Flow.** *Biochimica et Biophysica Acta (BBA) - General Subjects* **2014**, *1840*, 1492–1506, doi:10.1016/j.bbagen.2013.09.033.

1.2. AQP structural archetype: the core functional unit

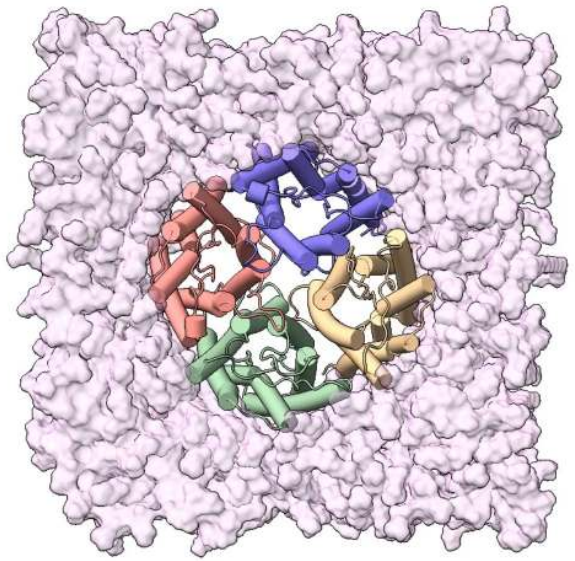


Human AQP4 tetramer in POPC bilayer

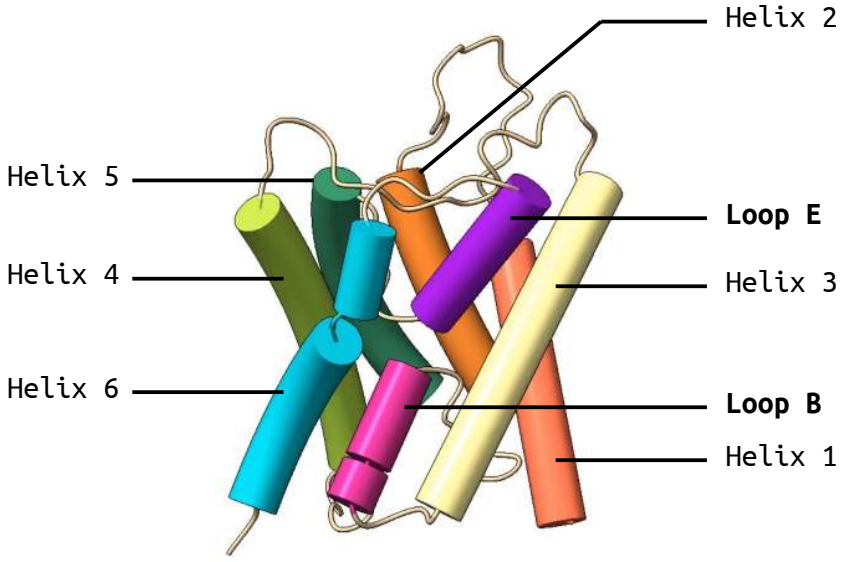


Human AQP4 monomer

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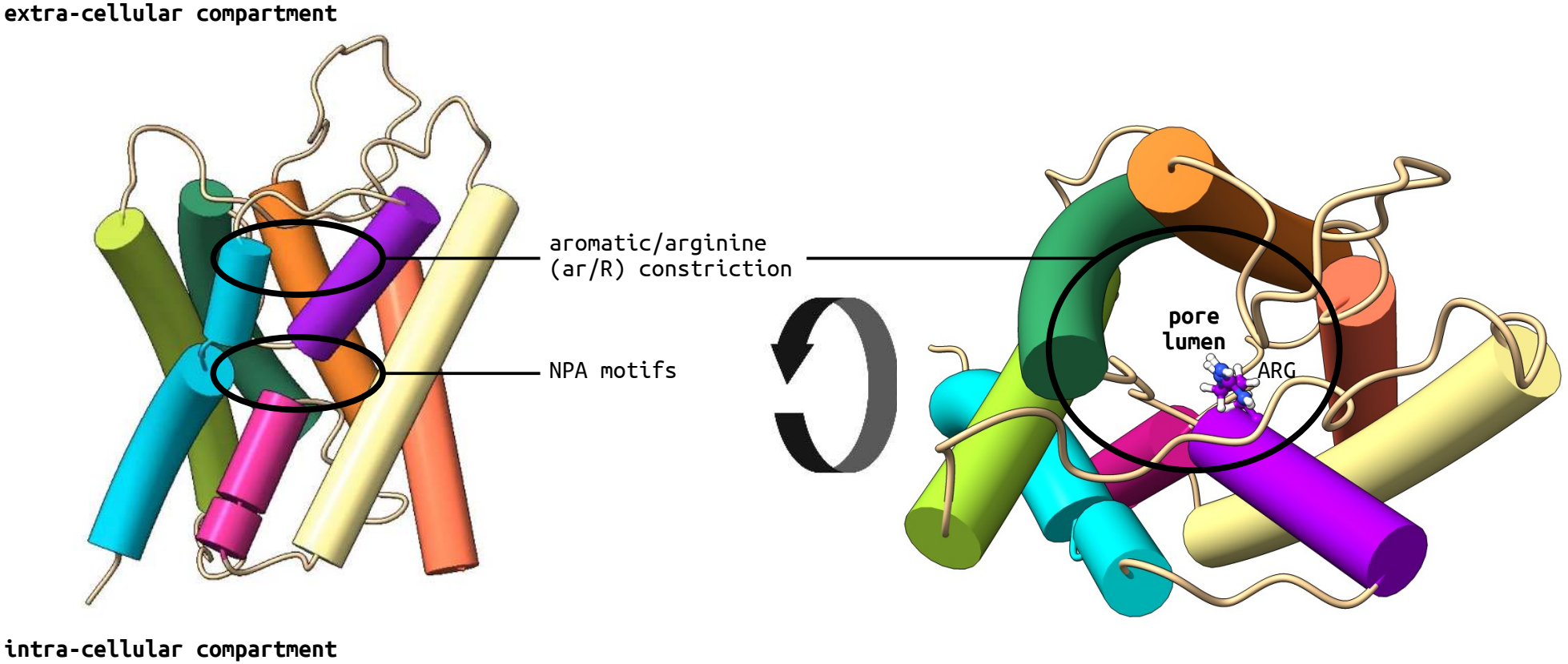


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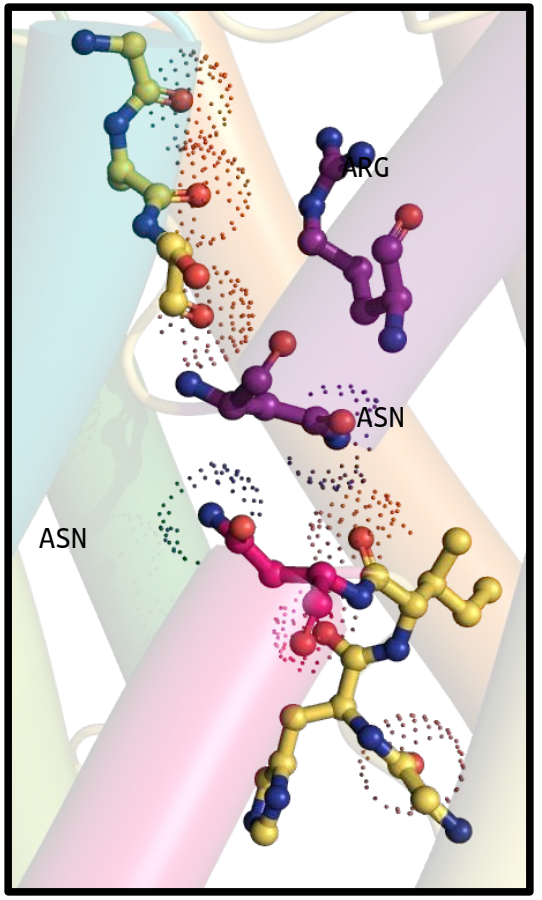


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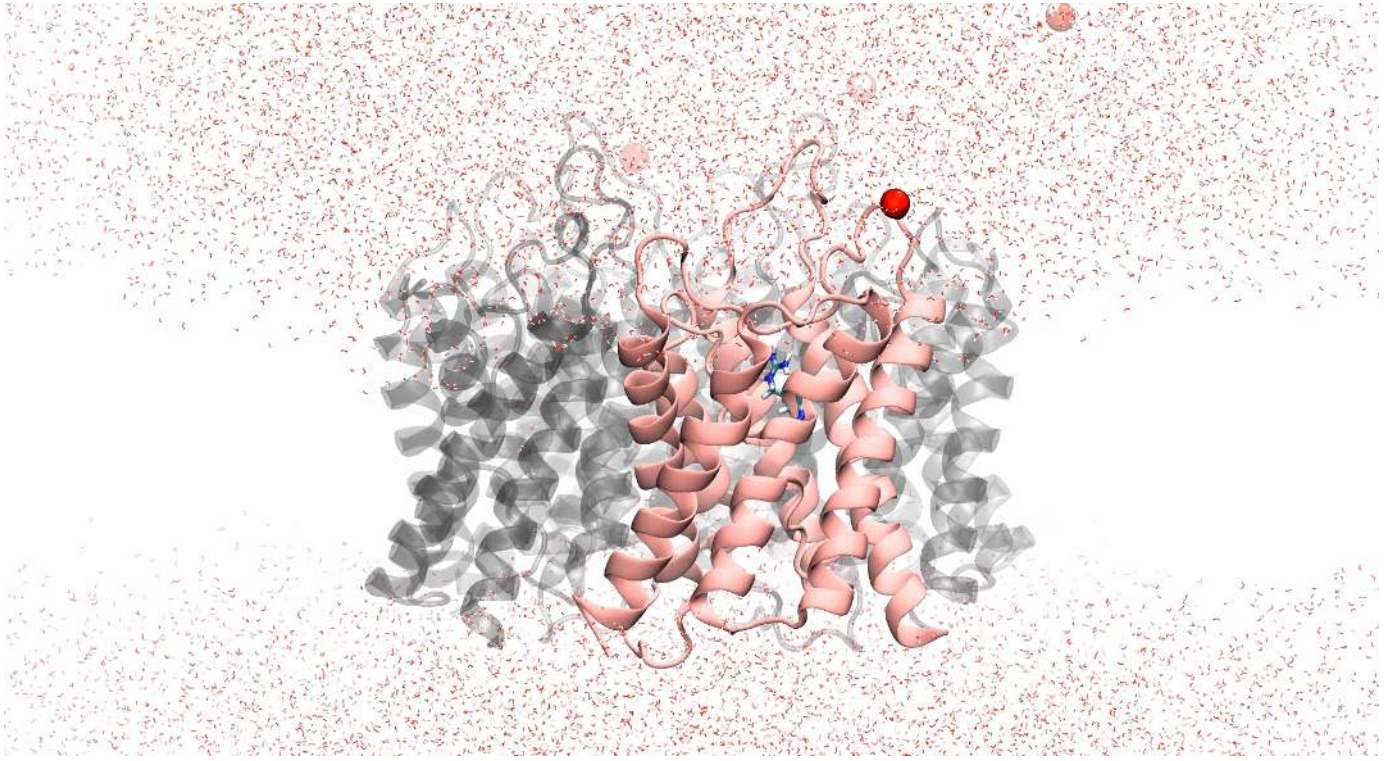
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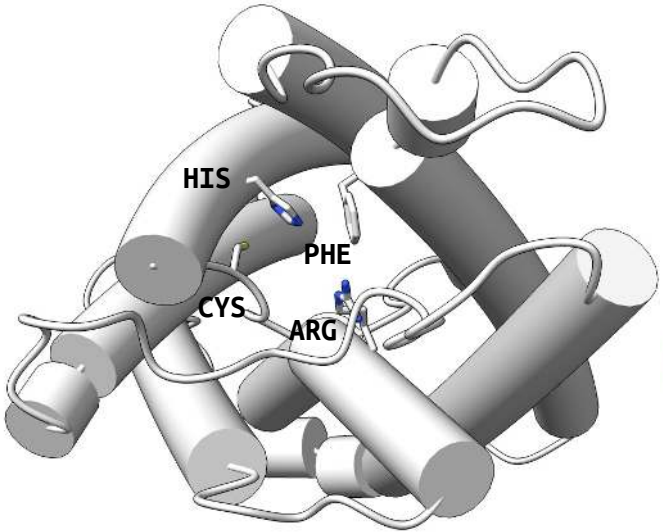


Hydrogen bonds relays are made by pore lining residues

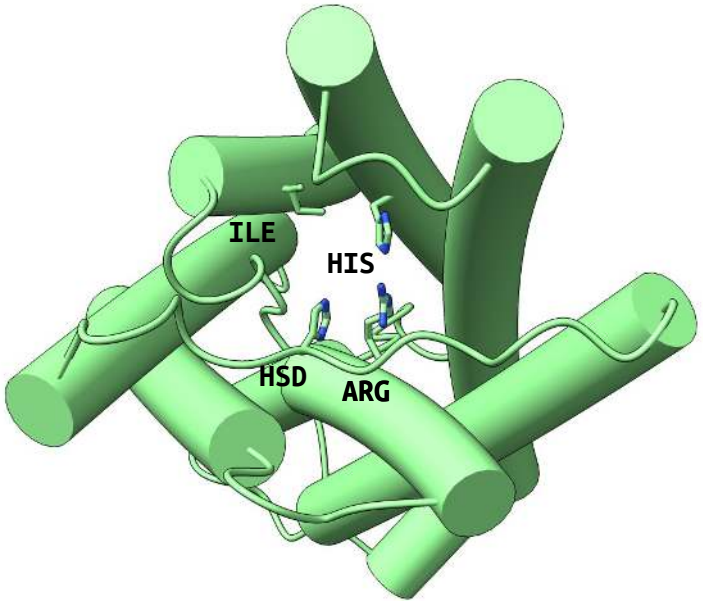


10 ns simulation of human AQP4 tetramer in POPC

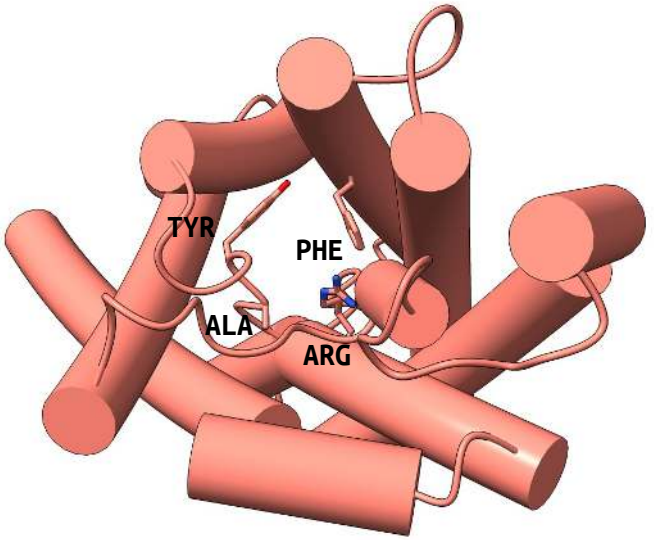
1.2. AQP structural archetype: the core functional unit



Strict AQP
(hAQP1, pdb 1j4n)

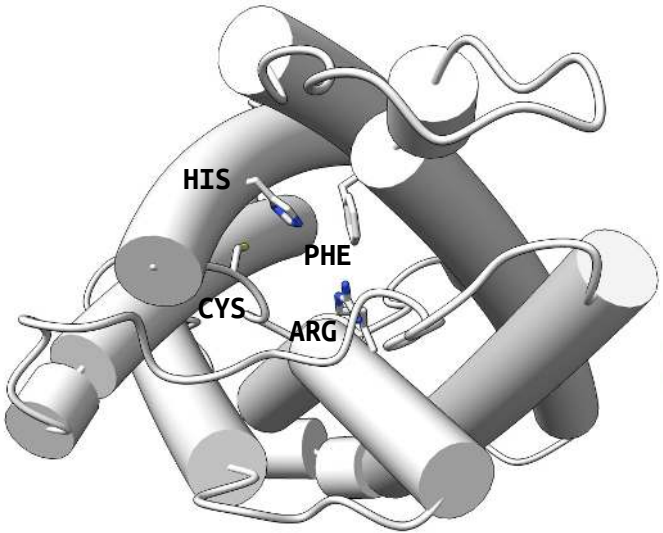


Aqua-ammoniaporin
(AtTIP2;1, pdb 5i32)

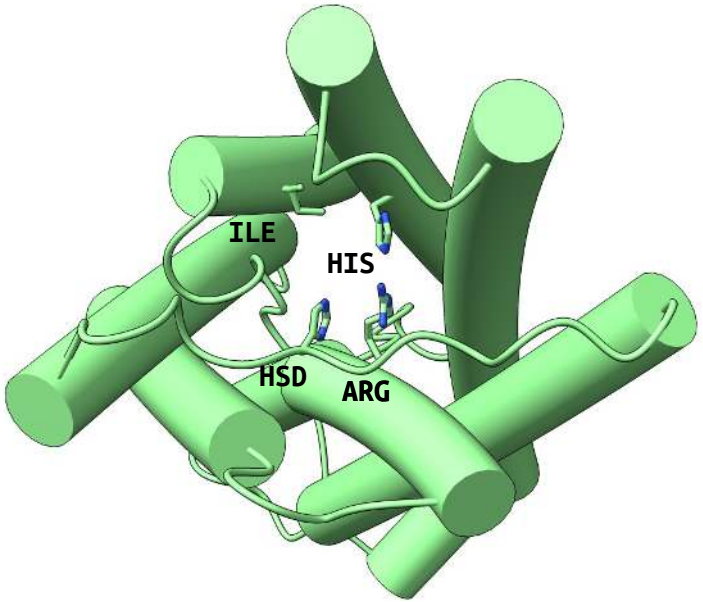


AQGP
(hAQP7, pdb 6qzi)

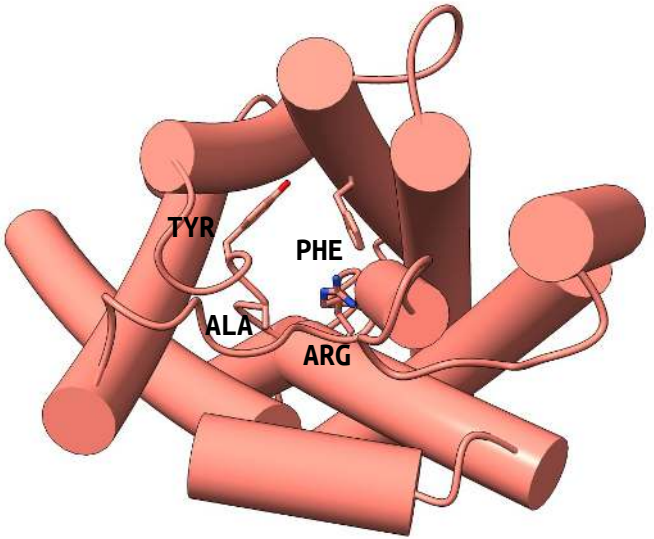
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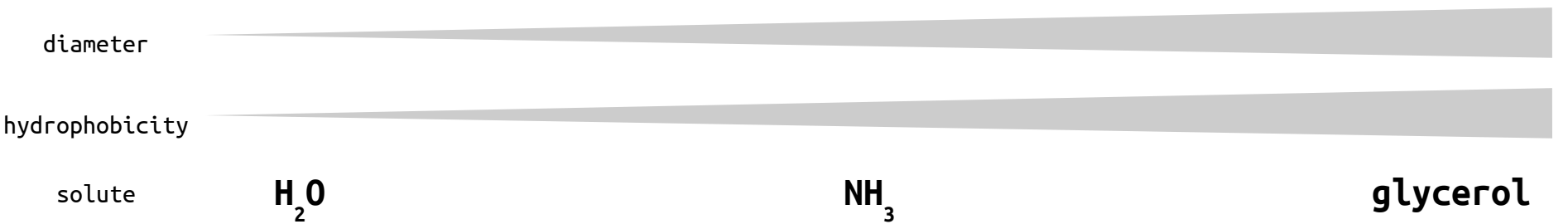
Strict AQP
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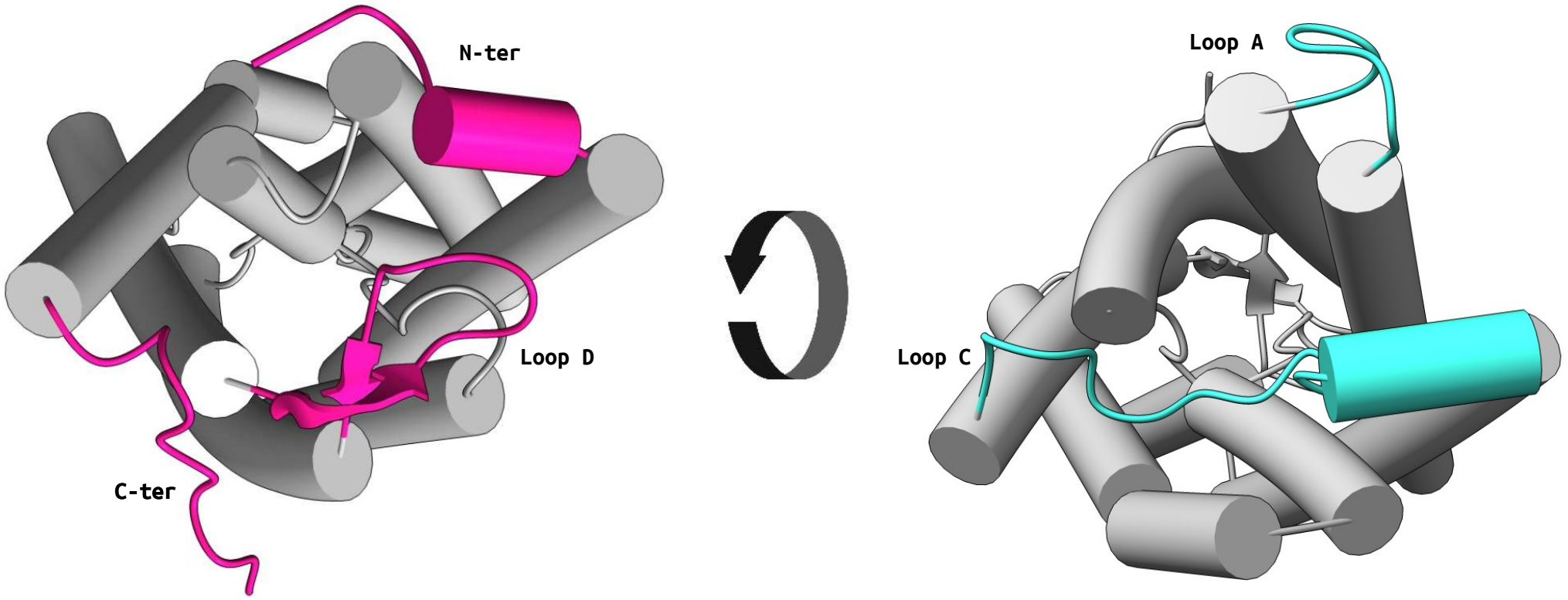
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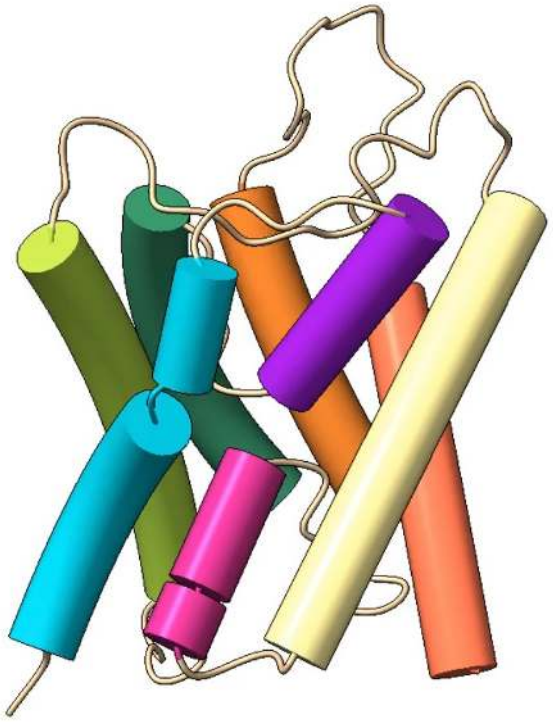


intra-cellular surface

SoPIP2;1
(pdb 1z98)

extra-cellular surface

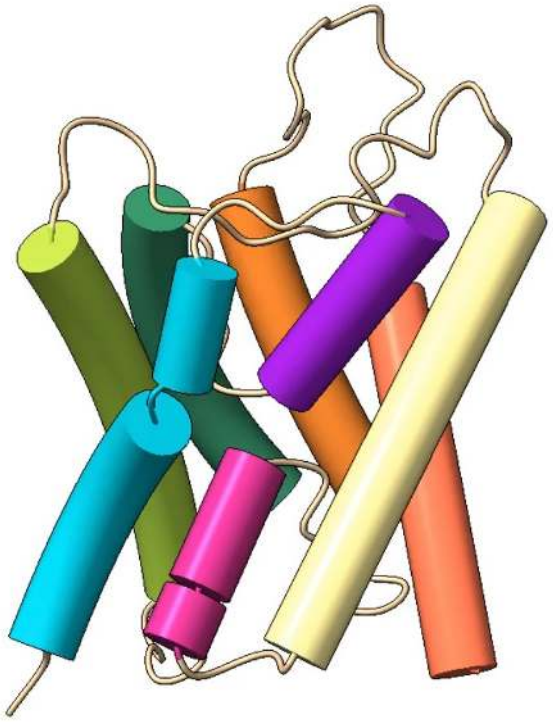
1.3. AQP structural archetype: simple design but high functional diversity



→ How can cells adjust their membrane permeability to all the fluctuations of their hydric environments / physiological roles ?

→ What is the function of the high variability of external loops and extremities ?

1.4. Molecular dynamics simulations



→ Molecular dynamics (MD) all atoms simulations

→ Newton’s second law: F = ma

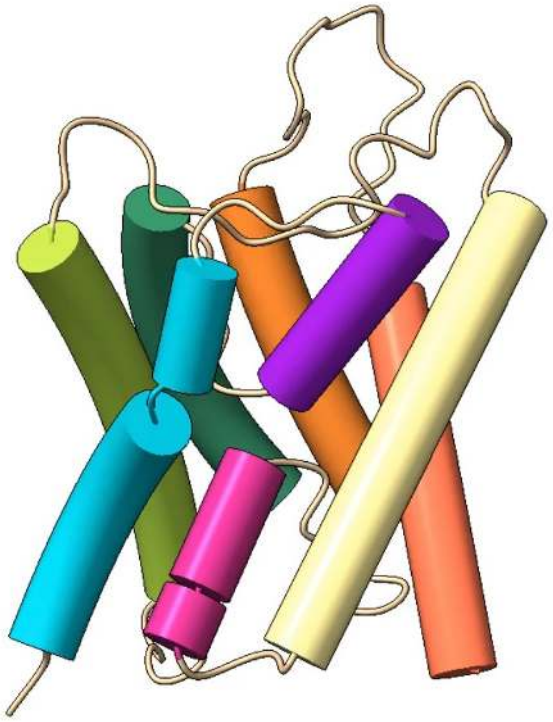
→ covalent bonds: harmonic and periodic potentials

→ Non-bounded interactions: Lennard-Jones and Coulomb potentials

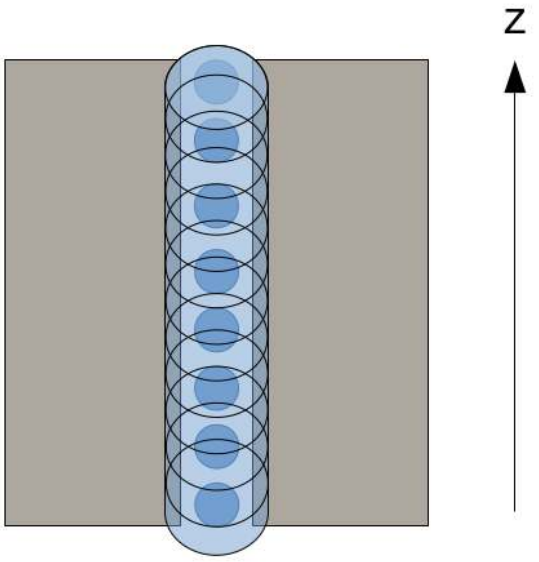
→ isothermal isobaric ensemble (NPT)

→ 310.15K / 1 bar / POPC bilayer / 150 mM KCl

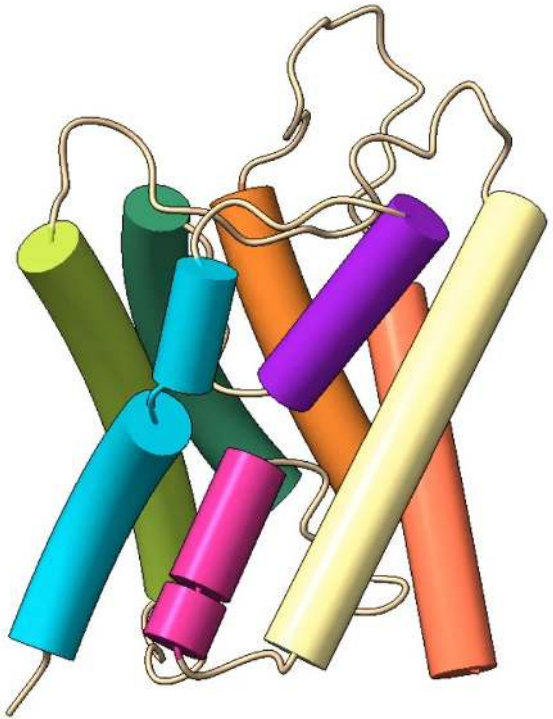
1.4. Molecular dynamics simulations



→ free energy profiles of water inside the conducting pore: $G(z) = -KT \ln(\rho(z)/\rho_{bulk})$



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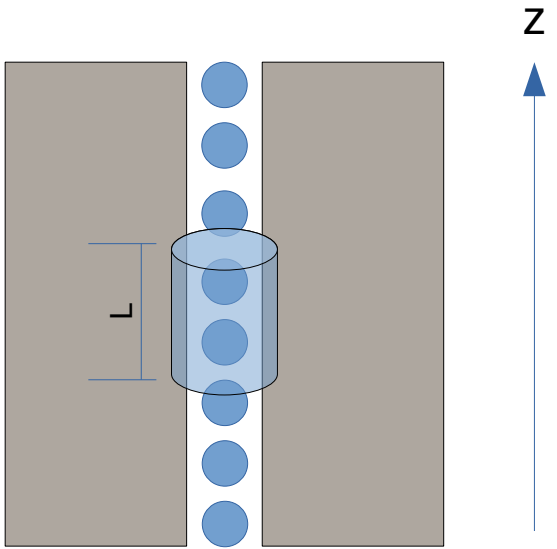
→ Water permeability coefficients:

$$j_W = p_f \Delta C_S$$

$$dn = \sum_{i \in S(t)} dz_i / L$$

$$\langle n^2(t) \rangle = 2 D_n t$$

$$p_f = \upsilon_W D_n$$



Zhu, F.; Tajkhorshid, E.; Schulten, K. **Collective Diffusion Model for Water Permeation through Microscopic Channels.** *Physical Review Letters* **2004**, *93*, 224501, doi:10.1103/PhysRevLett.93.224501.

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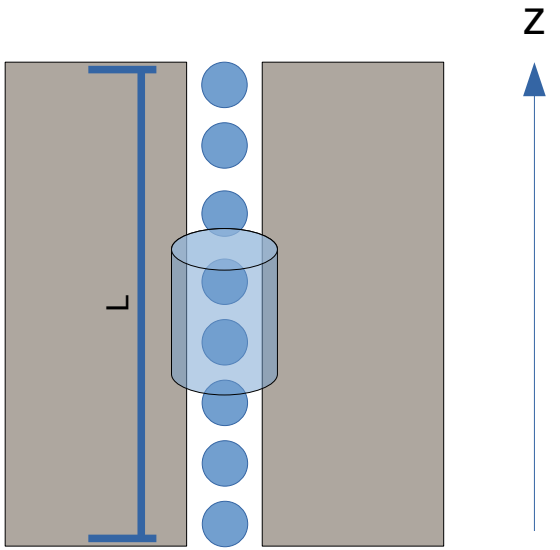
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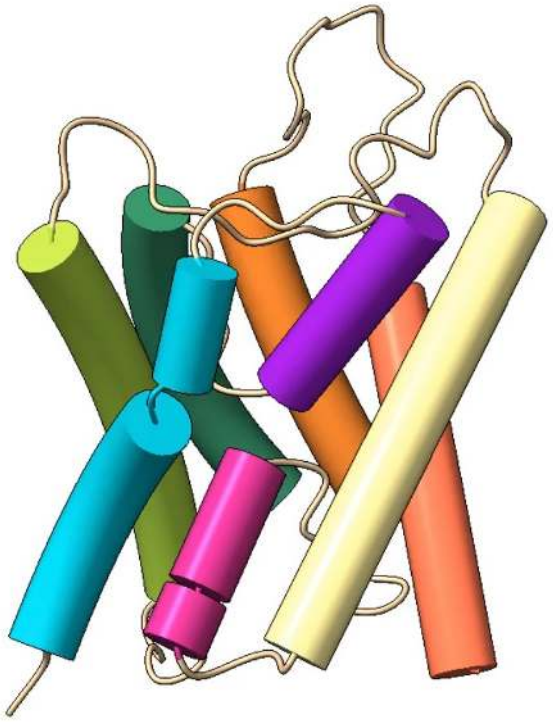
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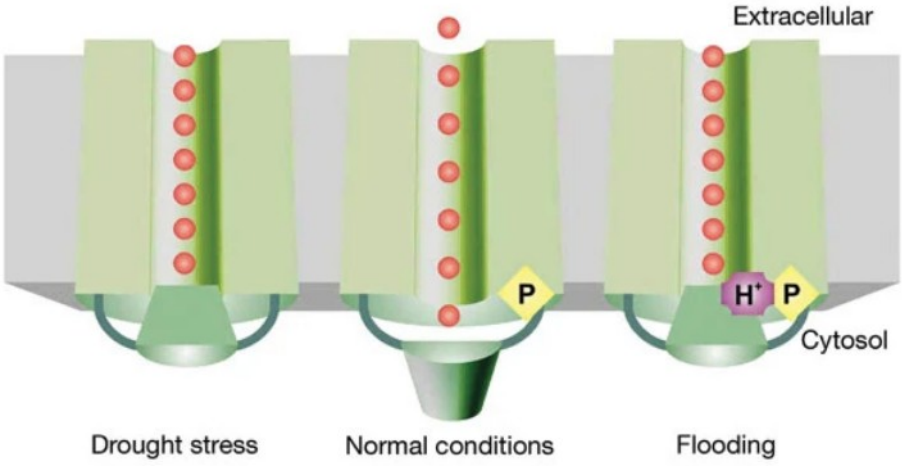


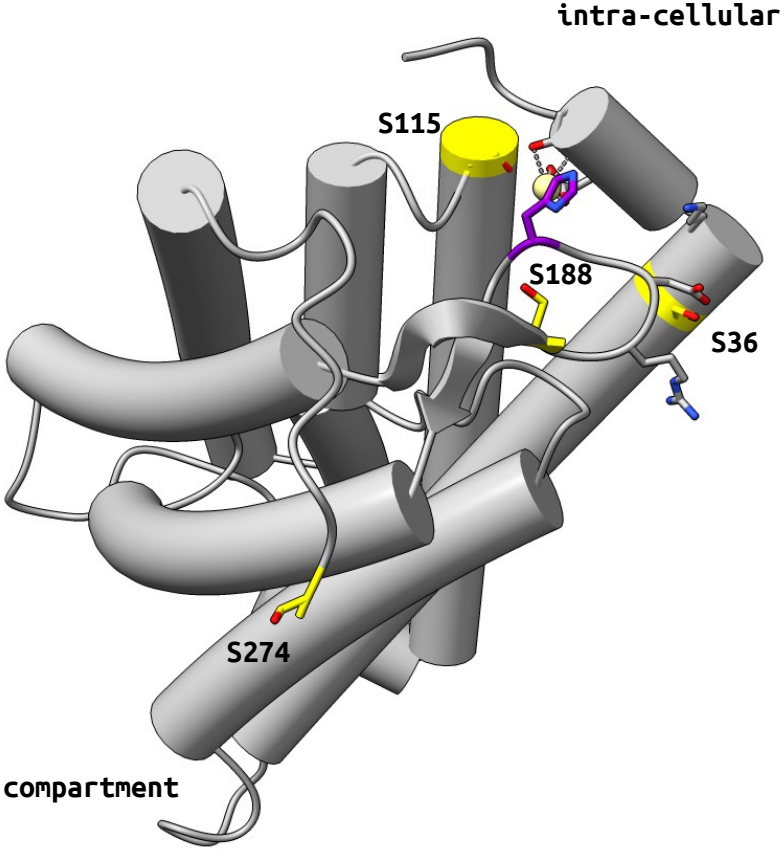
→ free energy profiles of water inside the conducting pore: $G(z) = -KT \ln(\rho(z)/\rho_{bulk})$

→ Water permeability coefficients

→ all trajectories were divided in 10ns sub-trajectories for statistical analysis

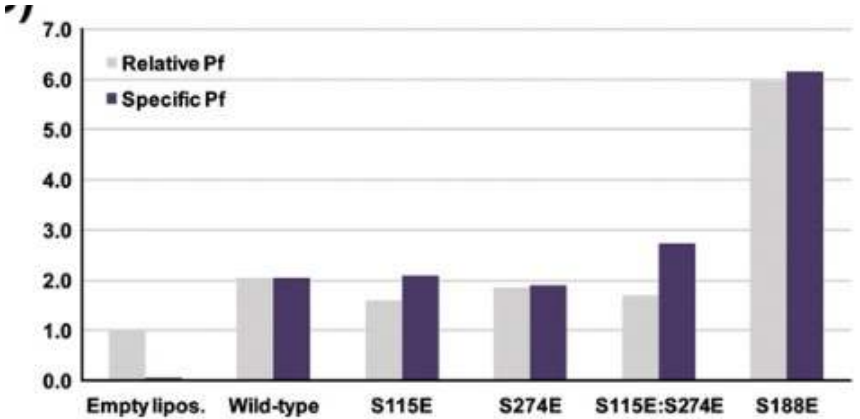
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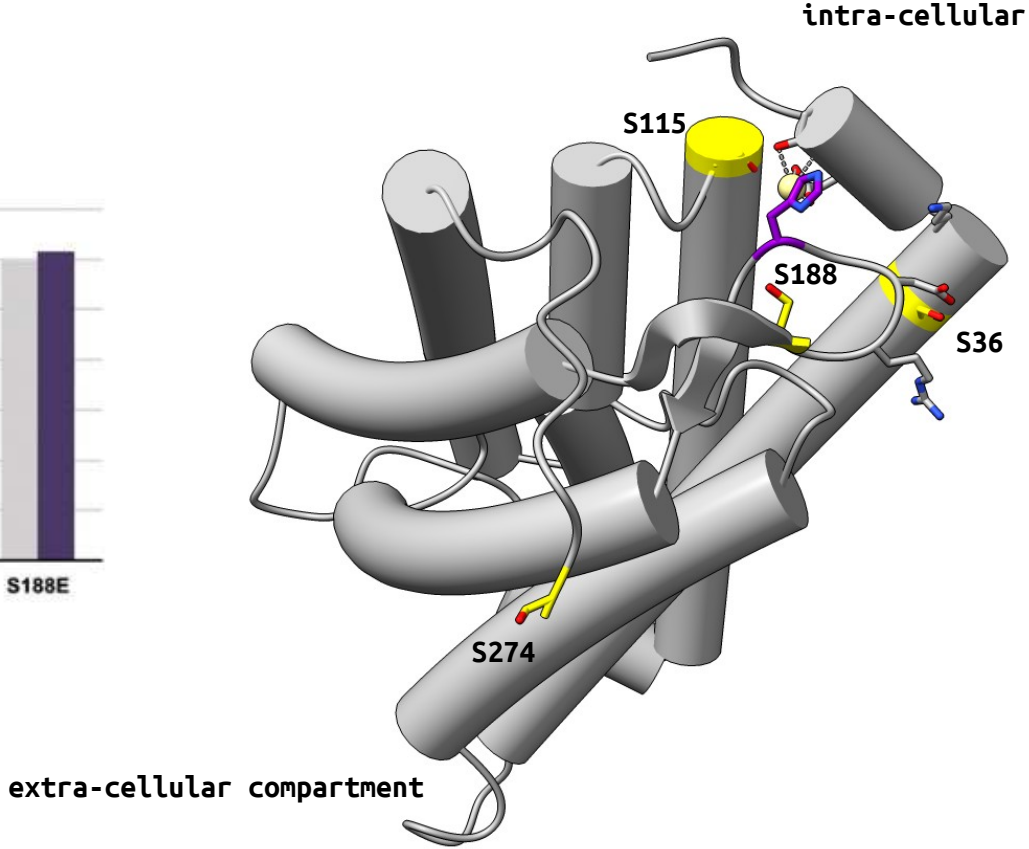




Törnroth-Horsefield, S.; Wang, Y.; Hedfalk, K.; Johanson, U.; Karlsson, M.; Tajkhorshid, E.; Neutze, R.; Kjellbom, P. **Structural Mechanism of Plant Aquaporin Gating.** *Nature* **2006**, *439*, 688–694, doi:10.1038/nature04316.

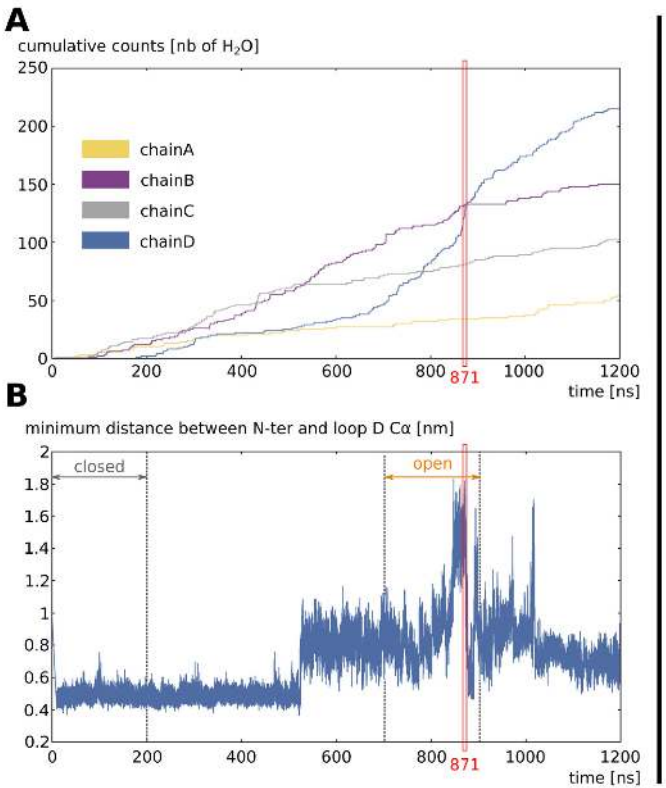
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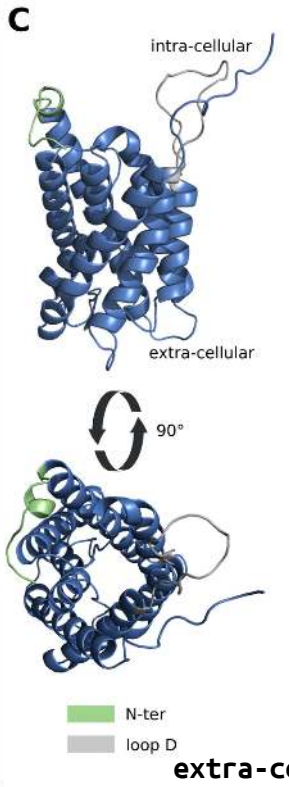


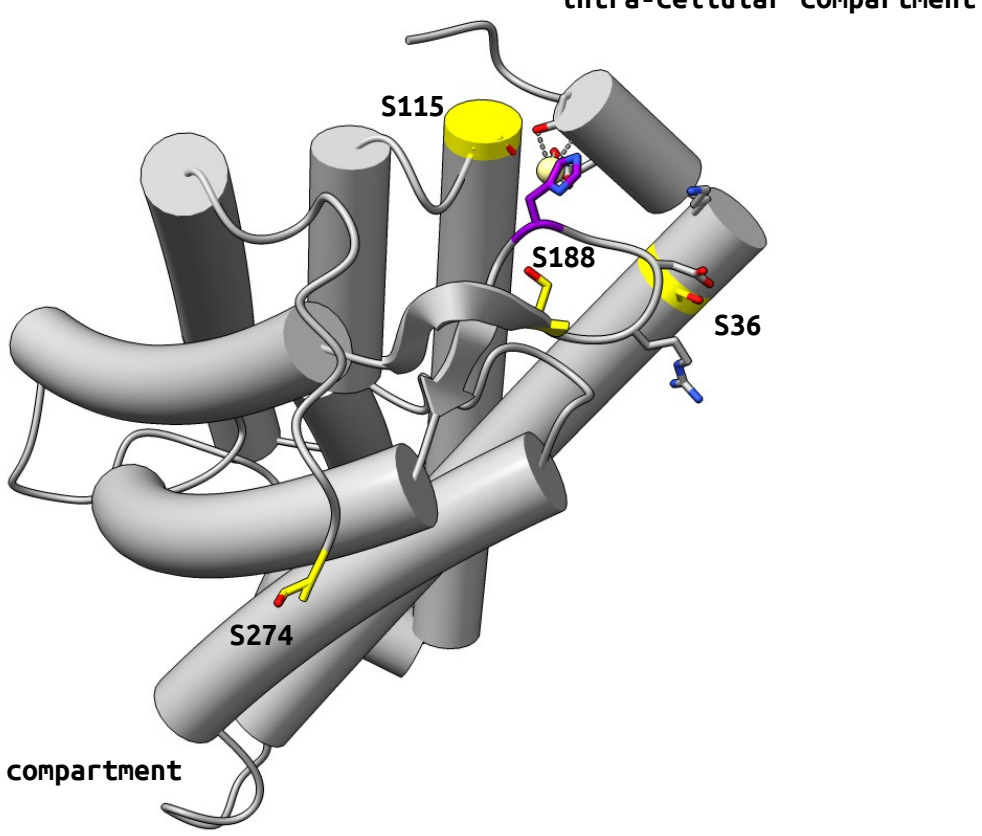


Nyblom, M.; Frick, A.; Wang, Y.; Ekvall, M.; Hallgren, K.; Hedfalk, K.; Neutze, R.; Tajkhorshid, E.; Törnroth-Horsefield, S. **Structural and Functional Analysis of SoPIP2;1 Mutants Adds Insight into Plant Aquaporin Gating. 2009**, *387*, 653–668, doi:10.1016/j.jmb.2009.01.065.

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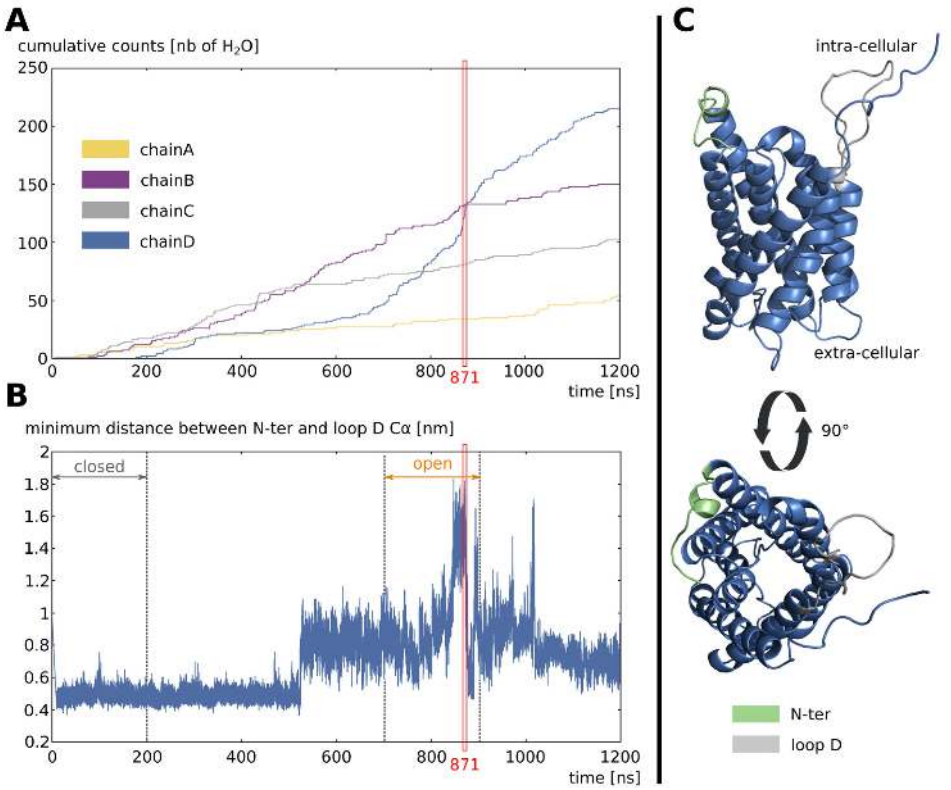


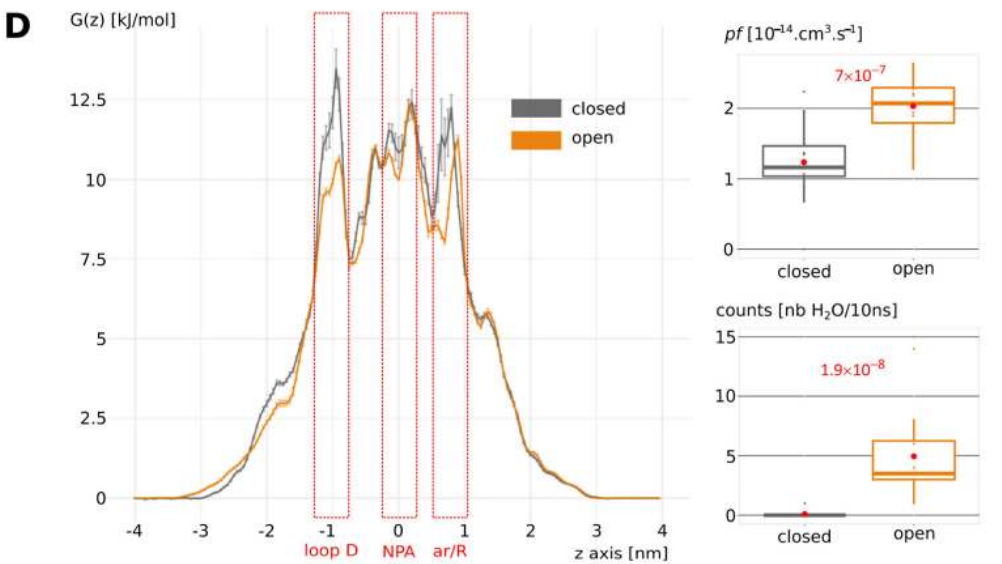




Mom, R.; Réty, S.; Mocquet, V.; Auguin, D. **Plant Aquaporin Gating Is Reversed by Phosphorylation on Intracellular Loop D—Evidence from Molecular Dynamics Simulations.** *International Journal of Molecular Sciences* **2023**, *24*, 13798, doi:10.3390/ijms241813798.

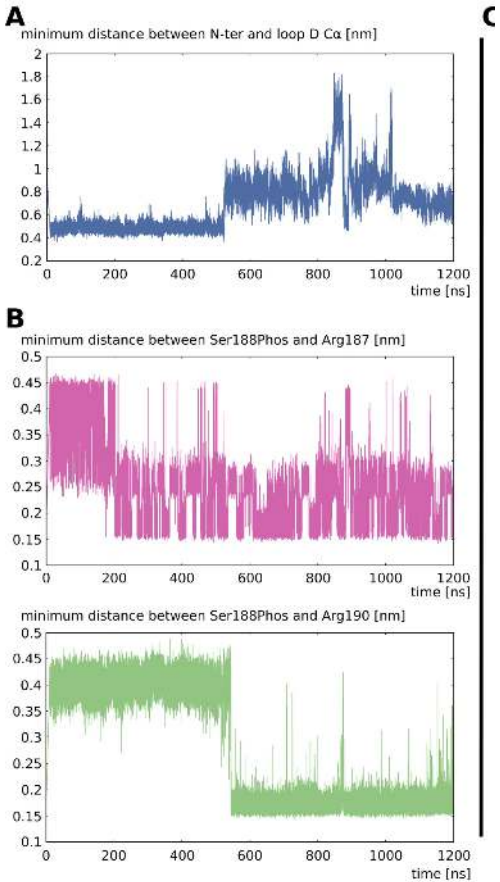
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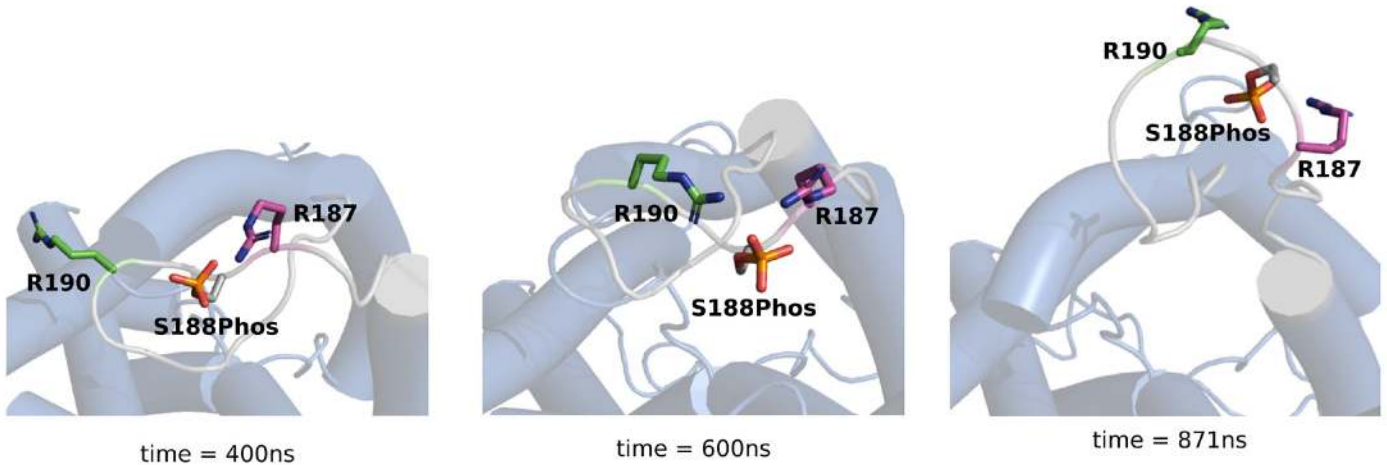




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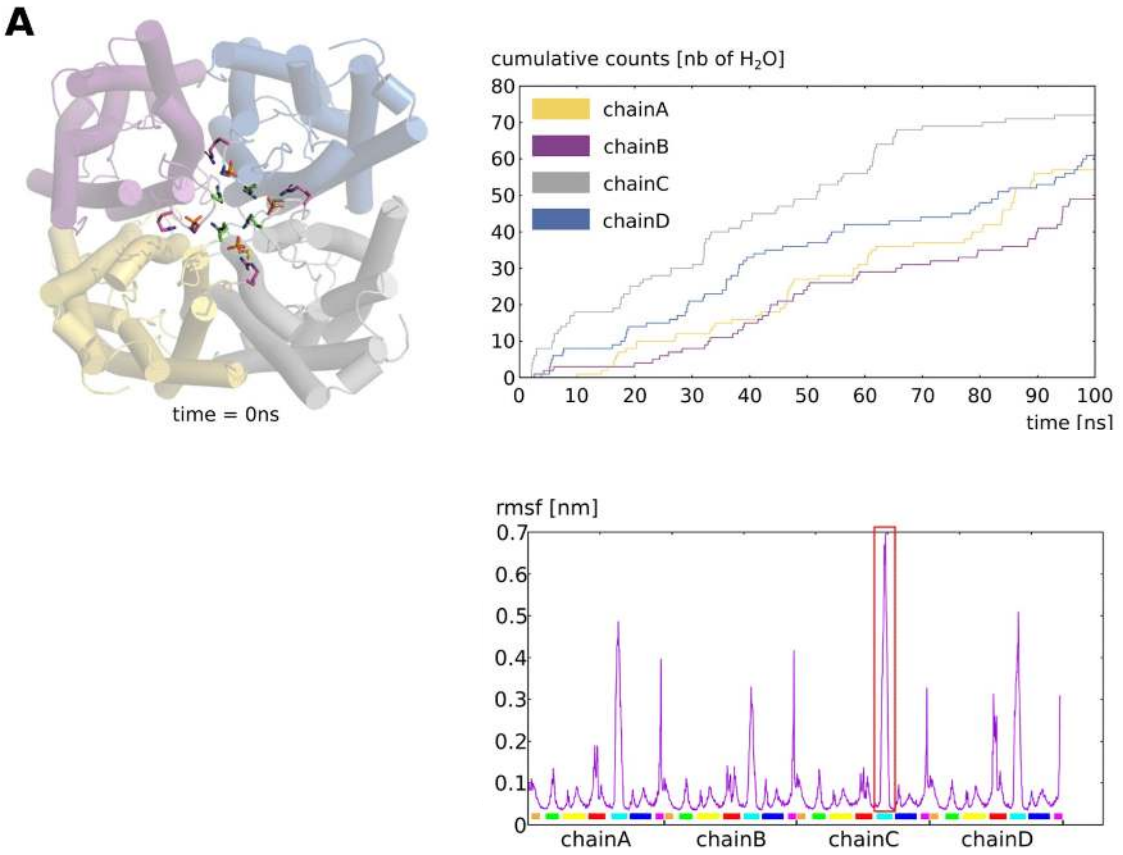
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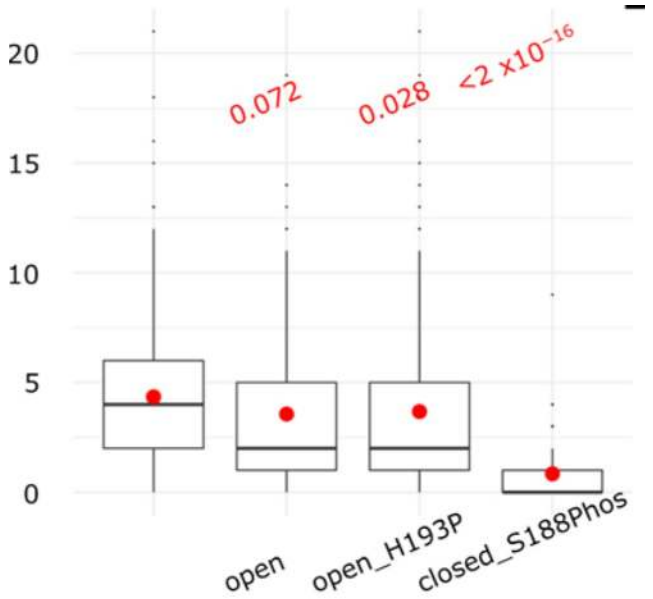
Mom, R.; Réty, S.; Mocquet, V.; Auguin, D. **Plant Aquaporin Gating Is Reversed by Phosphorylation on Intracellular Loop D—Evidence from Molecular Dynamics Simulations.** *International Journal of Molecular Sciences* **2023**, *24*, 13798, doi:10.3390/ijms241813798.

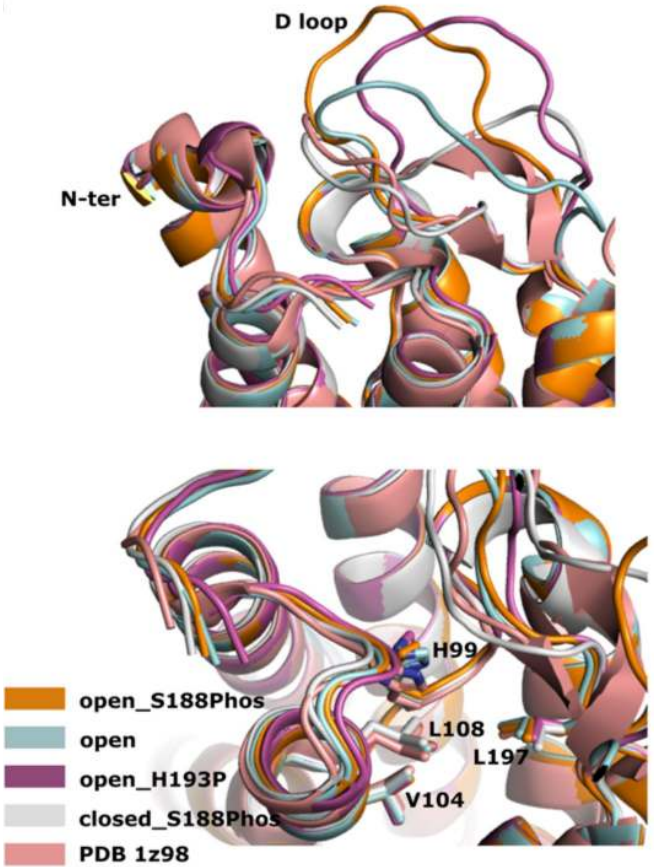
2.1. The plant case : a focus on intra-cellular gating



Mom, R.; Réty, S.; Mocquet, V.; Auguin, D. **Plant Aquaporin Gating Is Reversed by Phosphorylation on Intracellular Loop D—Evidence from Molecular Dynamics Simulations.** *International Journal of Molecular Sciences* **2023**, *24*, 13798, doi:10.3390/ijms241813798.

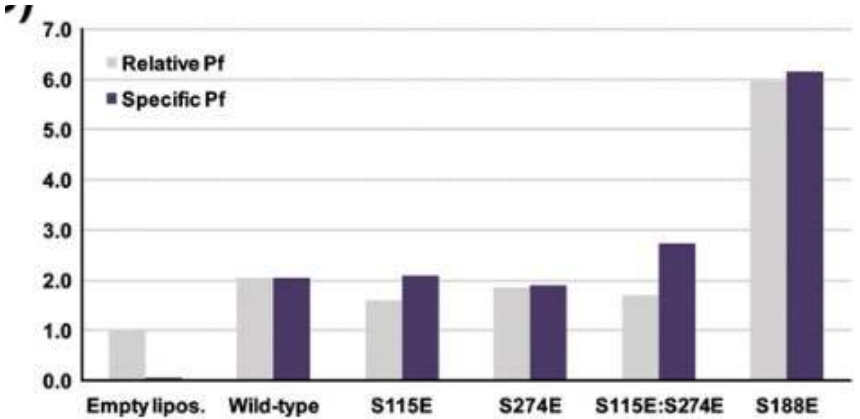
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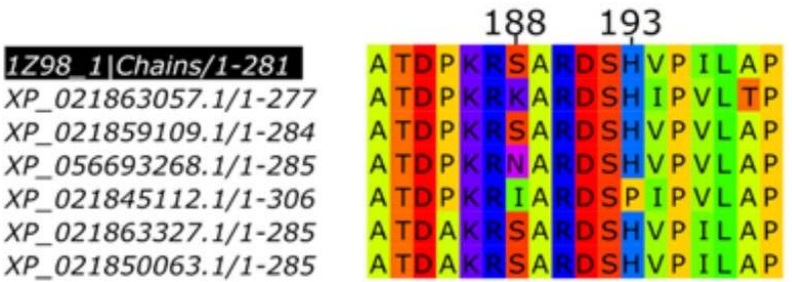




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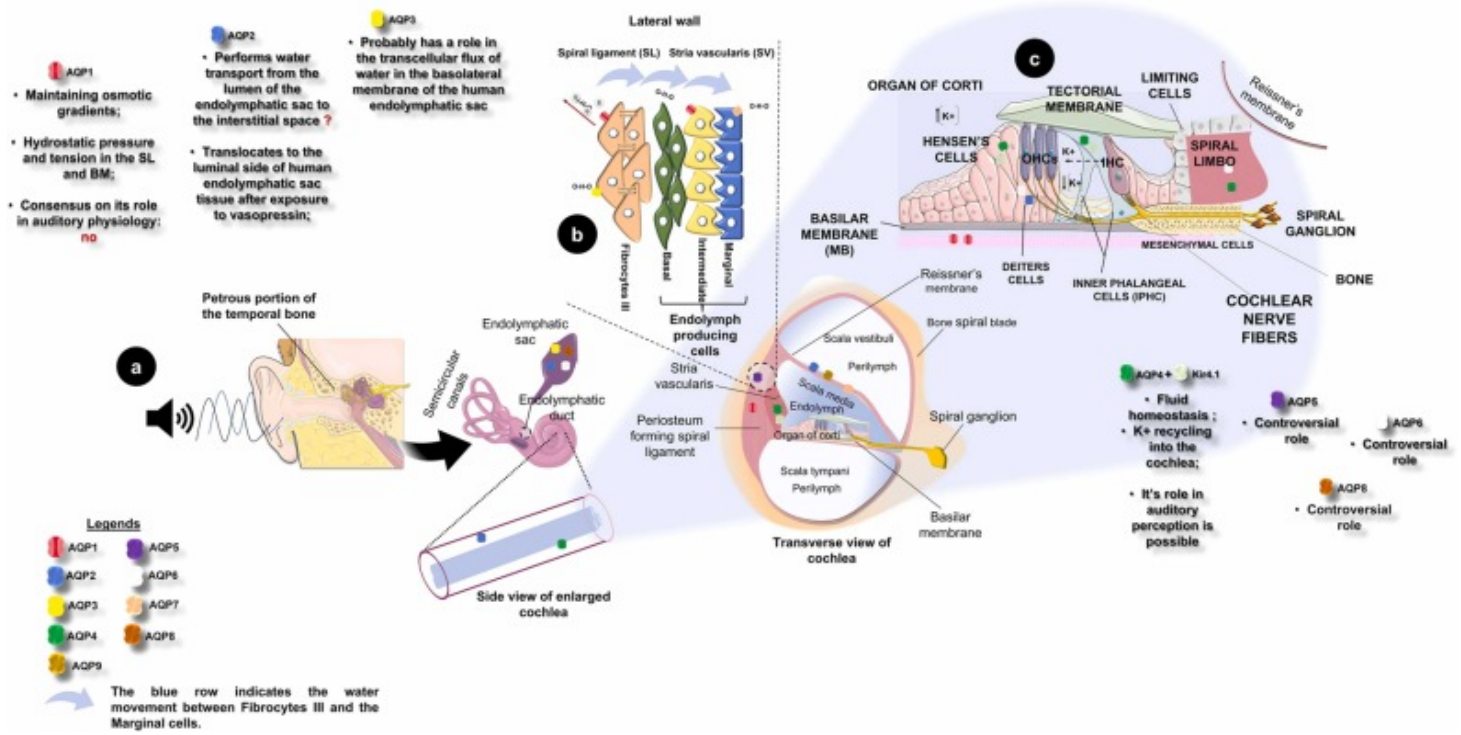
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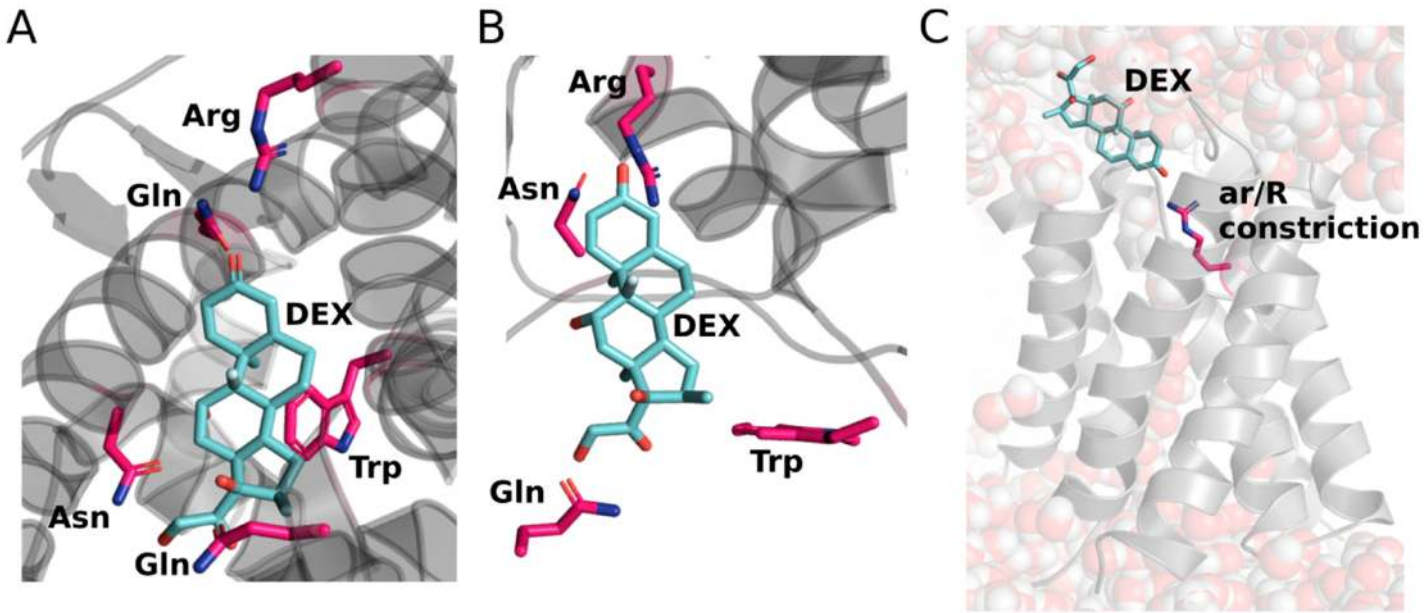
Nyblom, M.; Frick, A.; Wang, Y.; Ekvall, M.; Hallgren, K.; Hedfalk, K.; Neutze, R.; Tajkhorshid, E.; Törnroth-Horsefield, S. **Structural and Functional Analysis of SoPIP2;1 Mutants Adds Insight into Plant Aquaporin Gating. 2009**, *387*, 653–668, doi:10.1016/j.jmb.2009.01.065.

2.2. Modulation by steroids : the role of extra-cellular loops



Ximenes-da-Silva, A.; Capra, D.; Sanz, C.K.; Mendes, C.B.; de Mattos Coelho Aguiar, J.; Moura-Neto, V.; DosSantos, M.F. **The Role of Aquaporins in Hearing Function and Dysfunction.** *Eur J Cell Biol* **2022**, *101*, 151252, doi:10.1016/j.ejcb.2022.151252.

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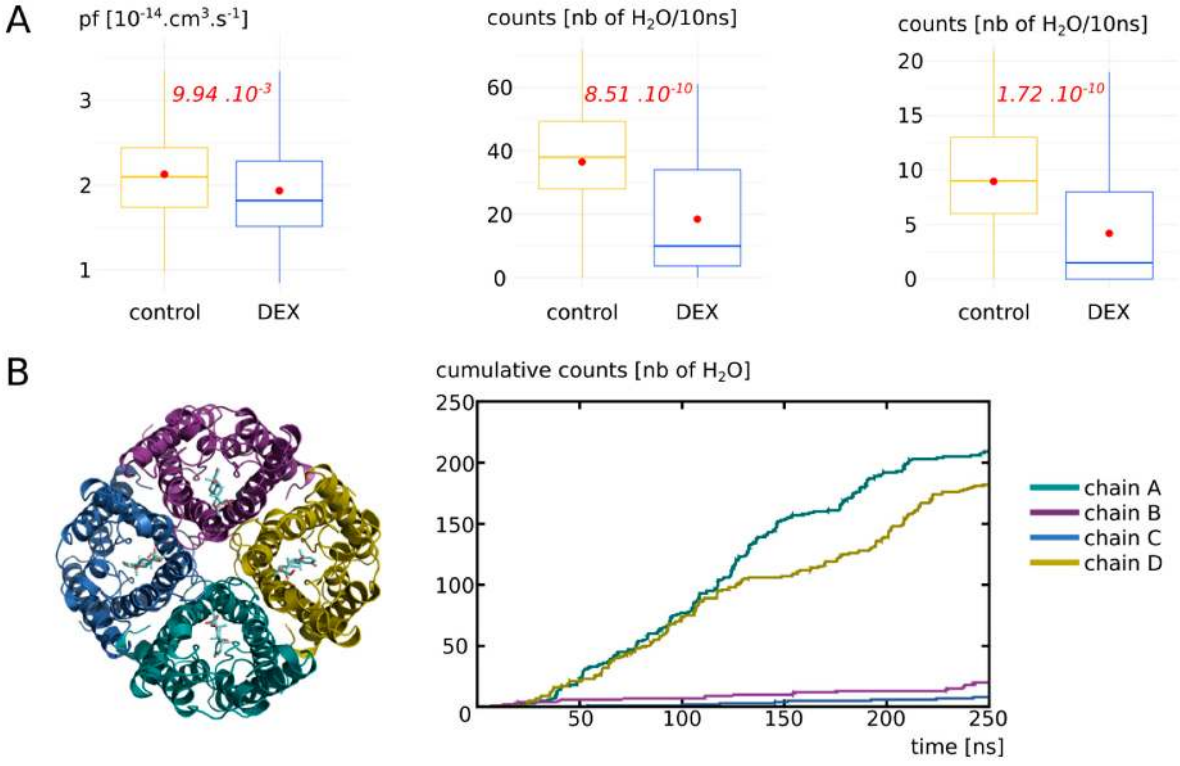


Glucocorticoid Receptor
(pdb, 3mne)

Human AQP2
(pdb, 4nef)

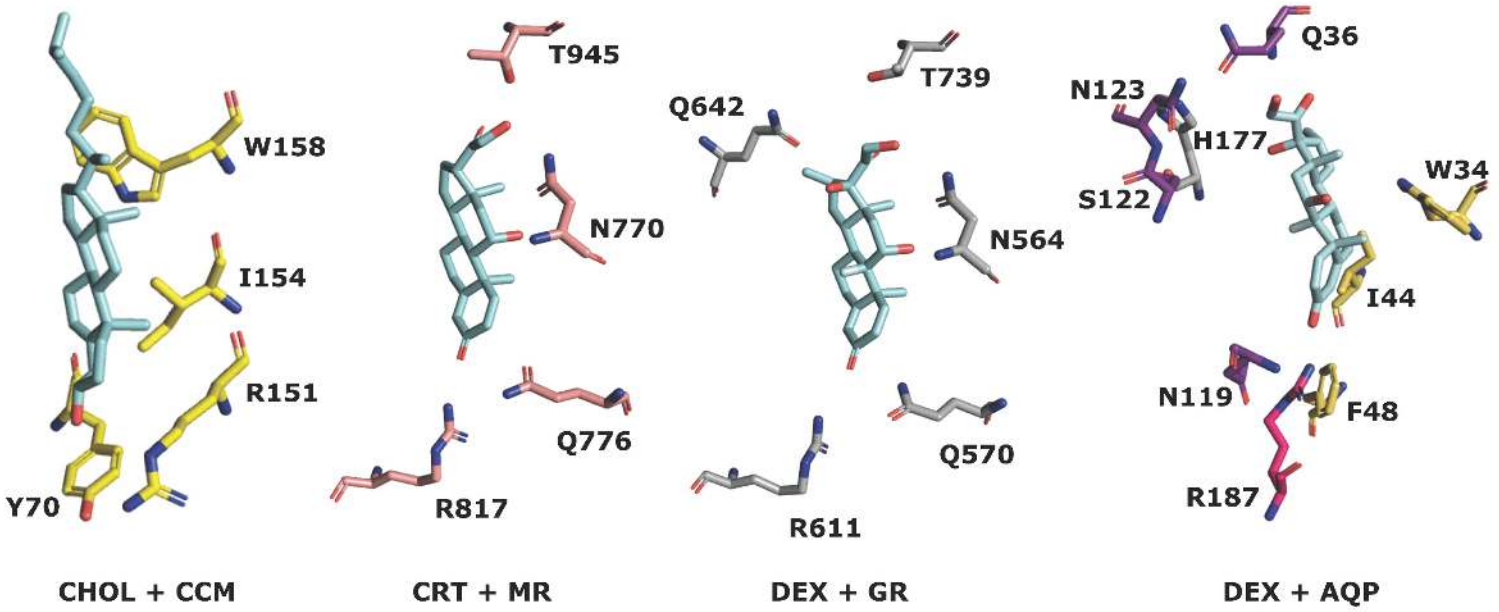
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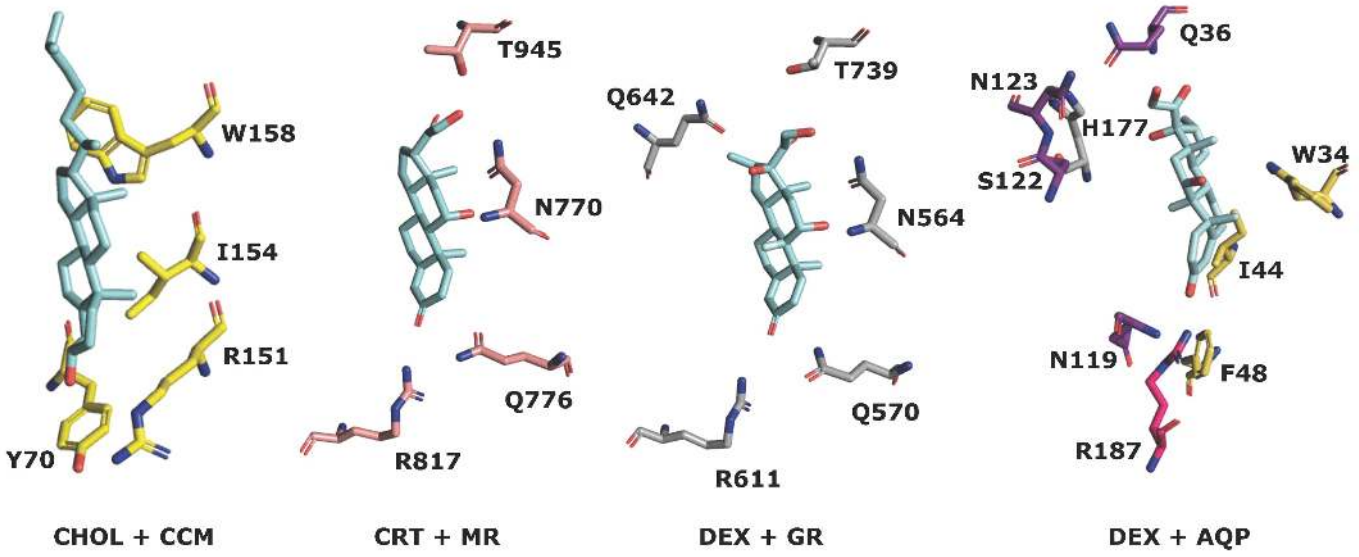
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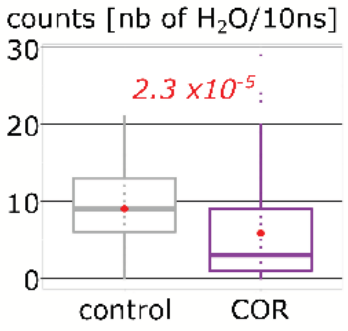
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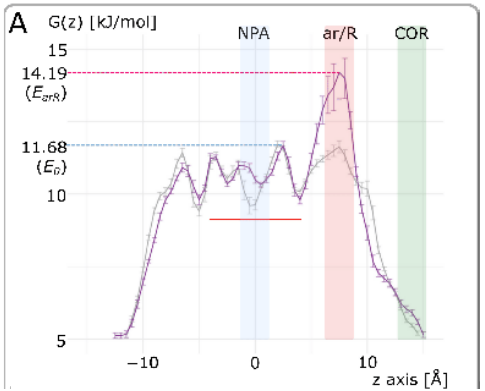
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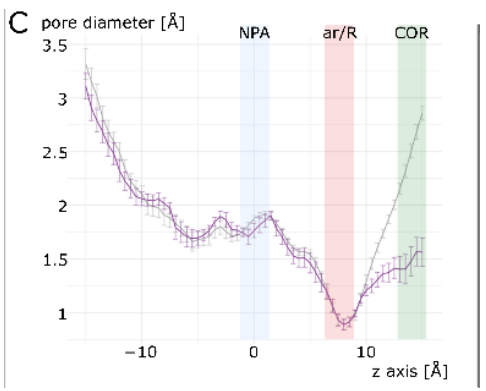


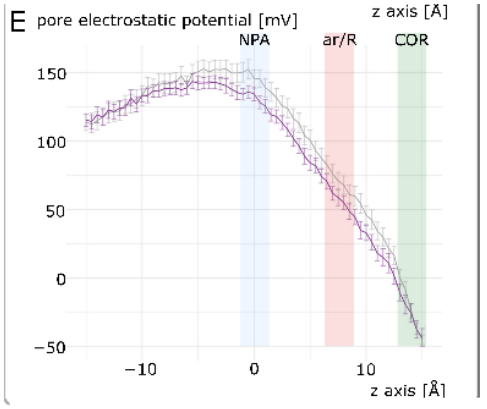


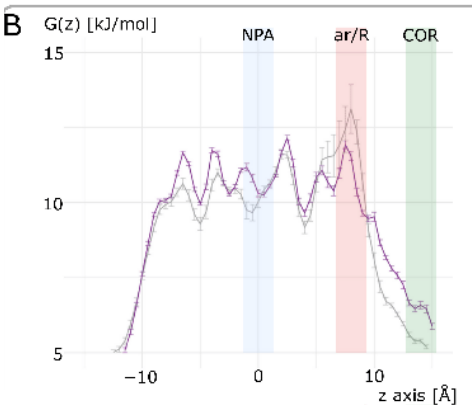
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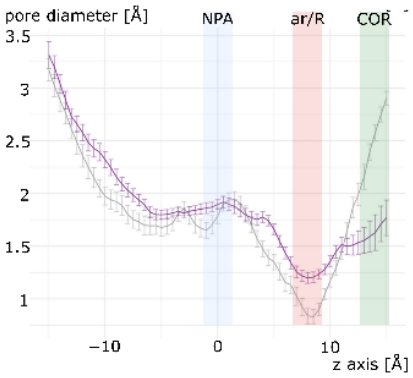
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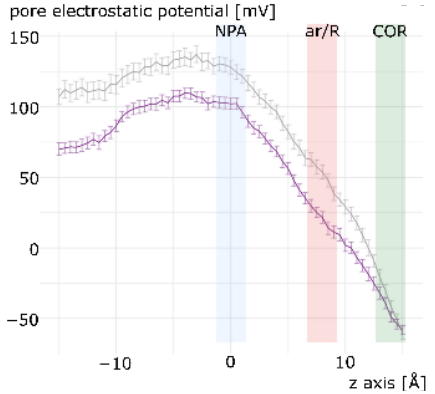






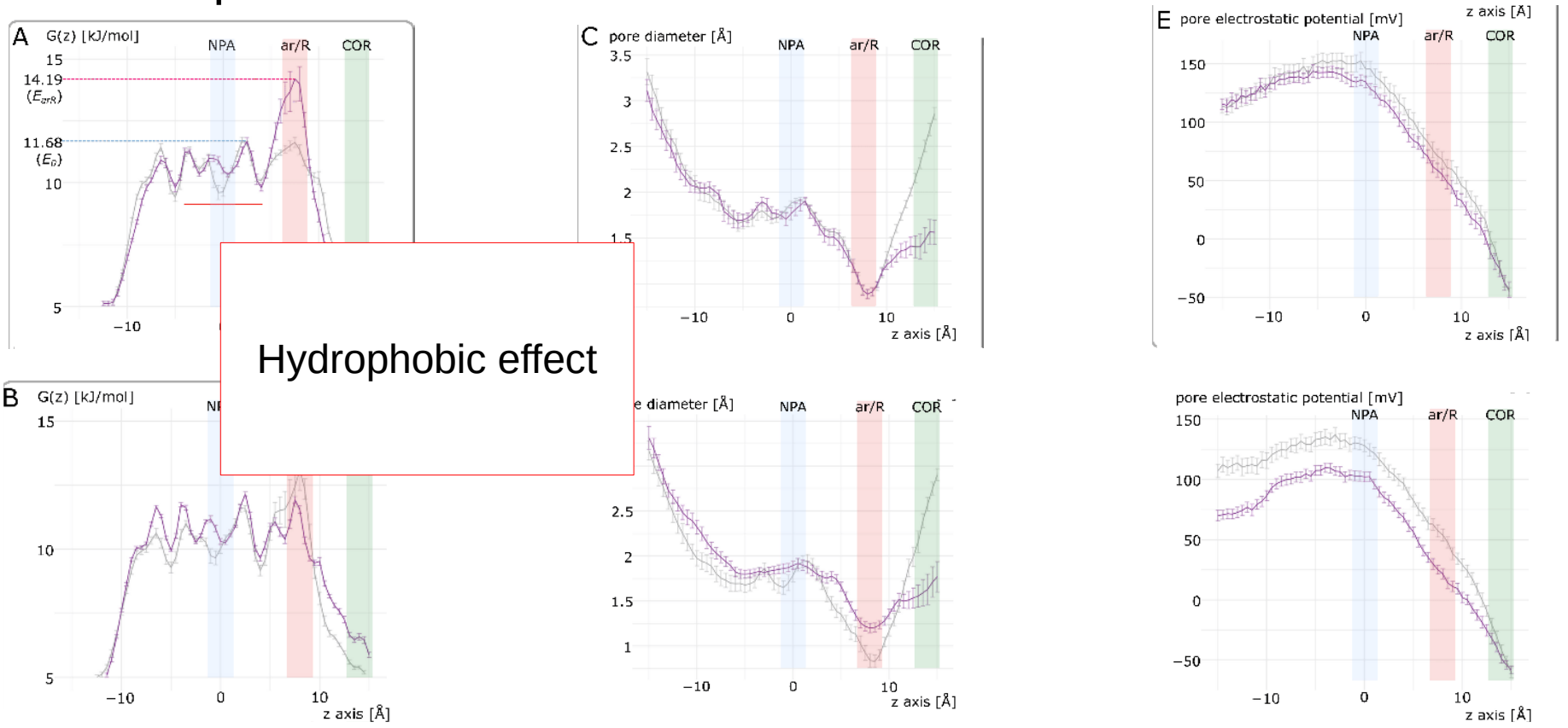






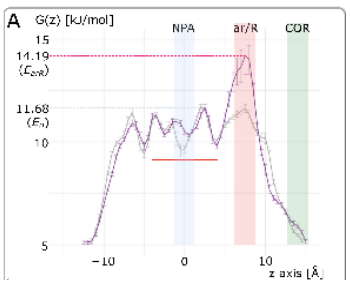
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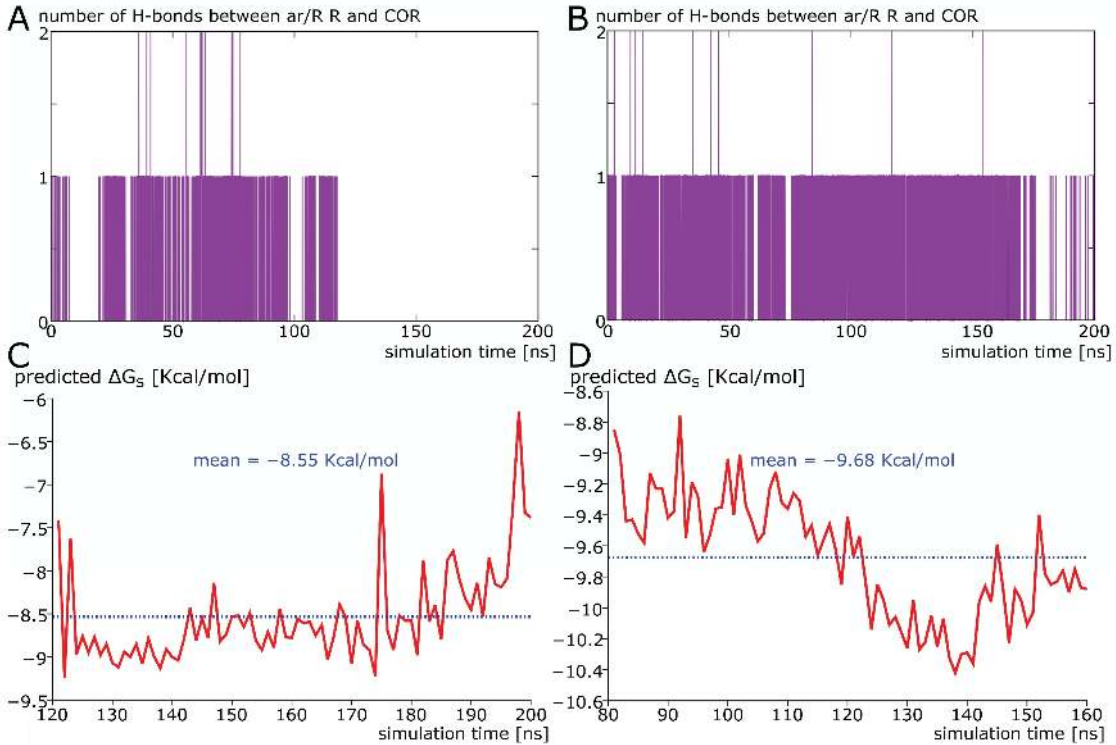
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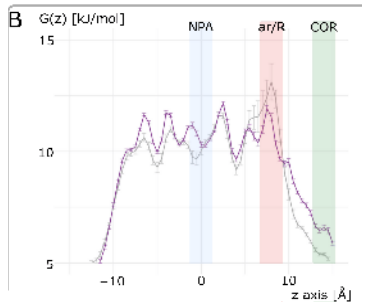


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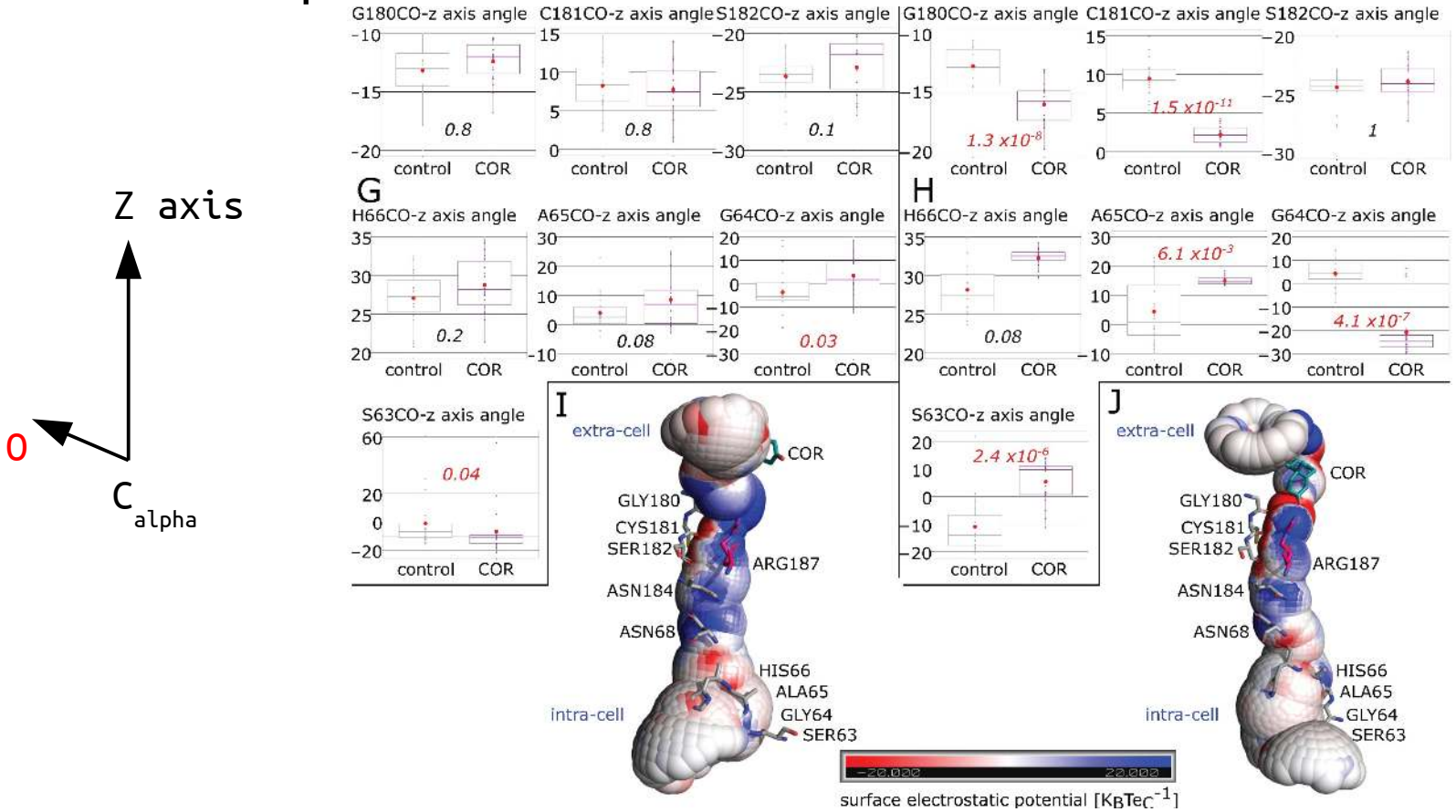






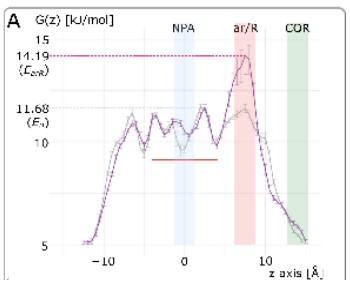
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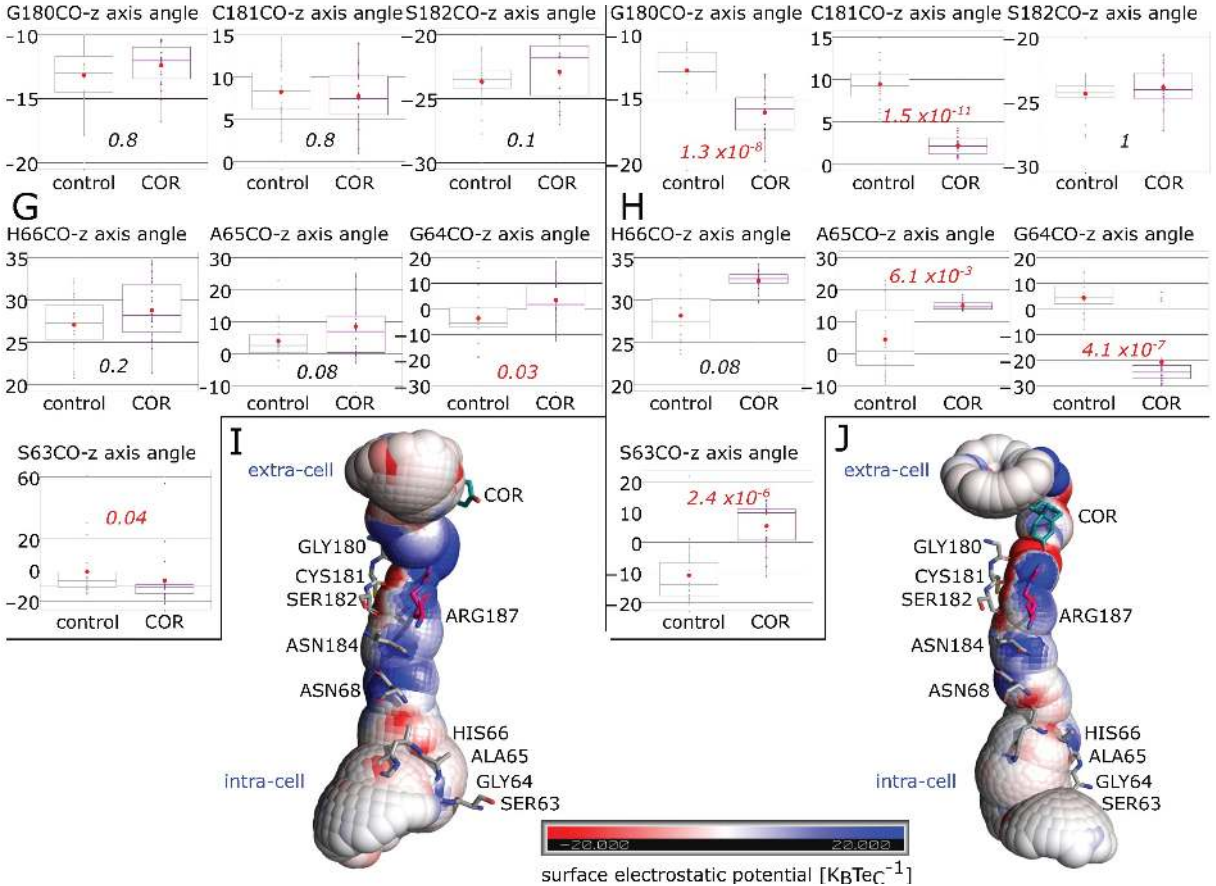
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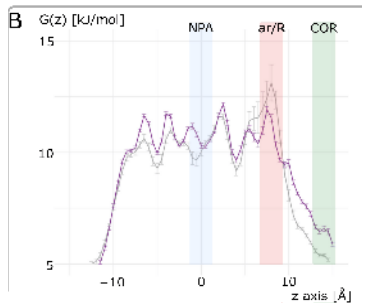


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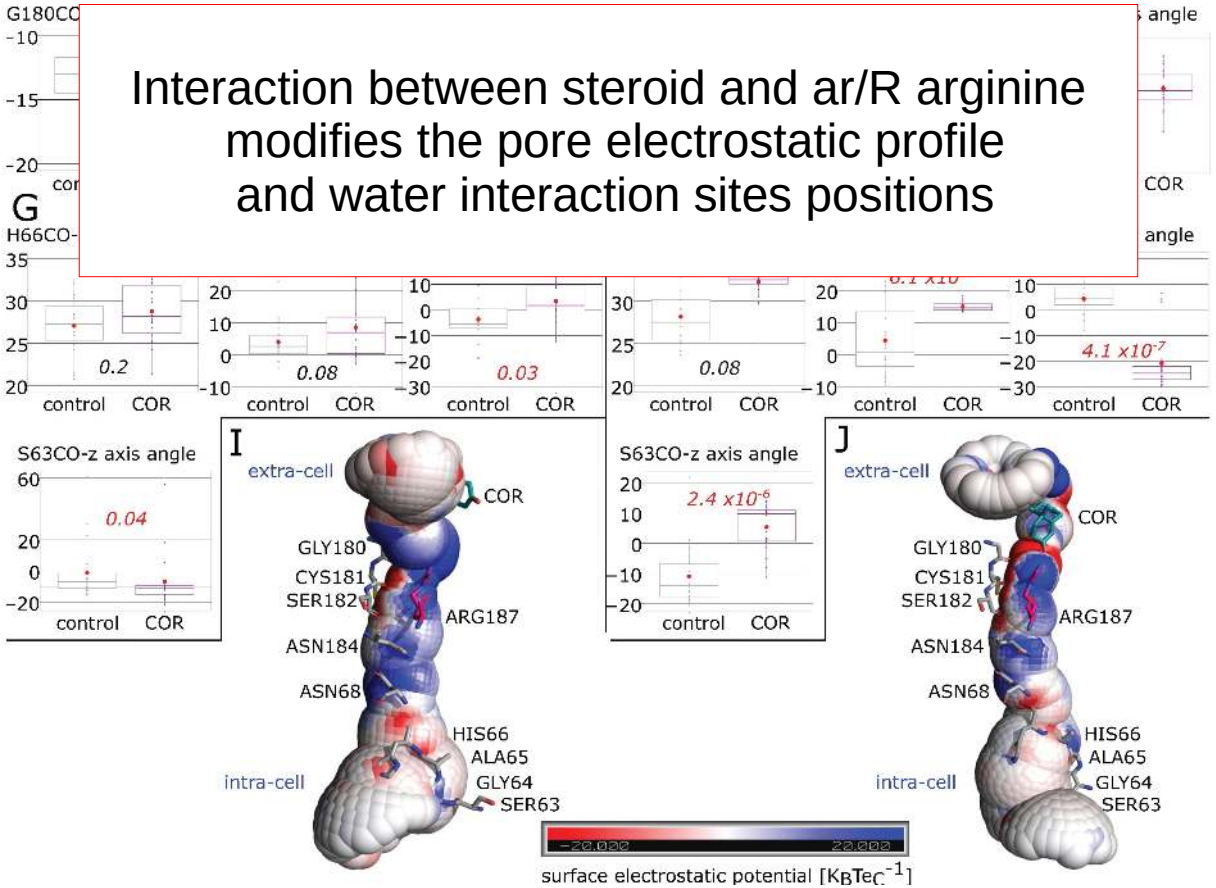






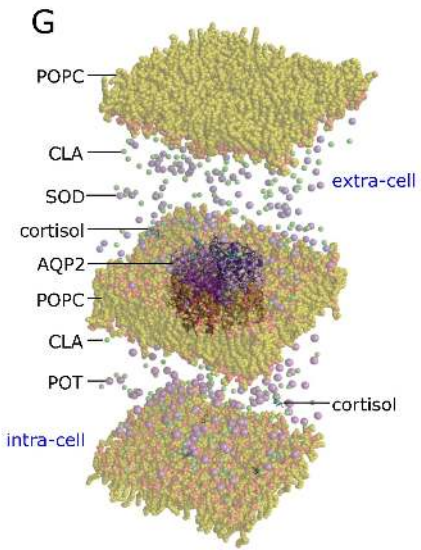
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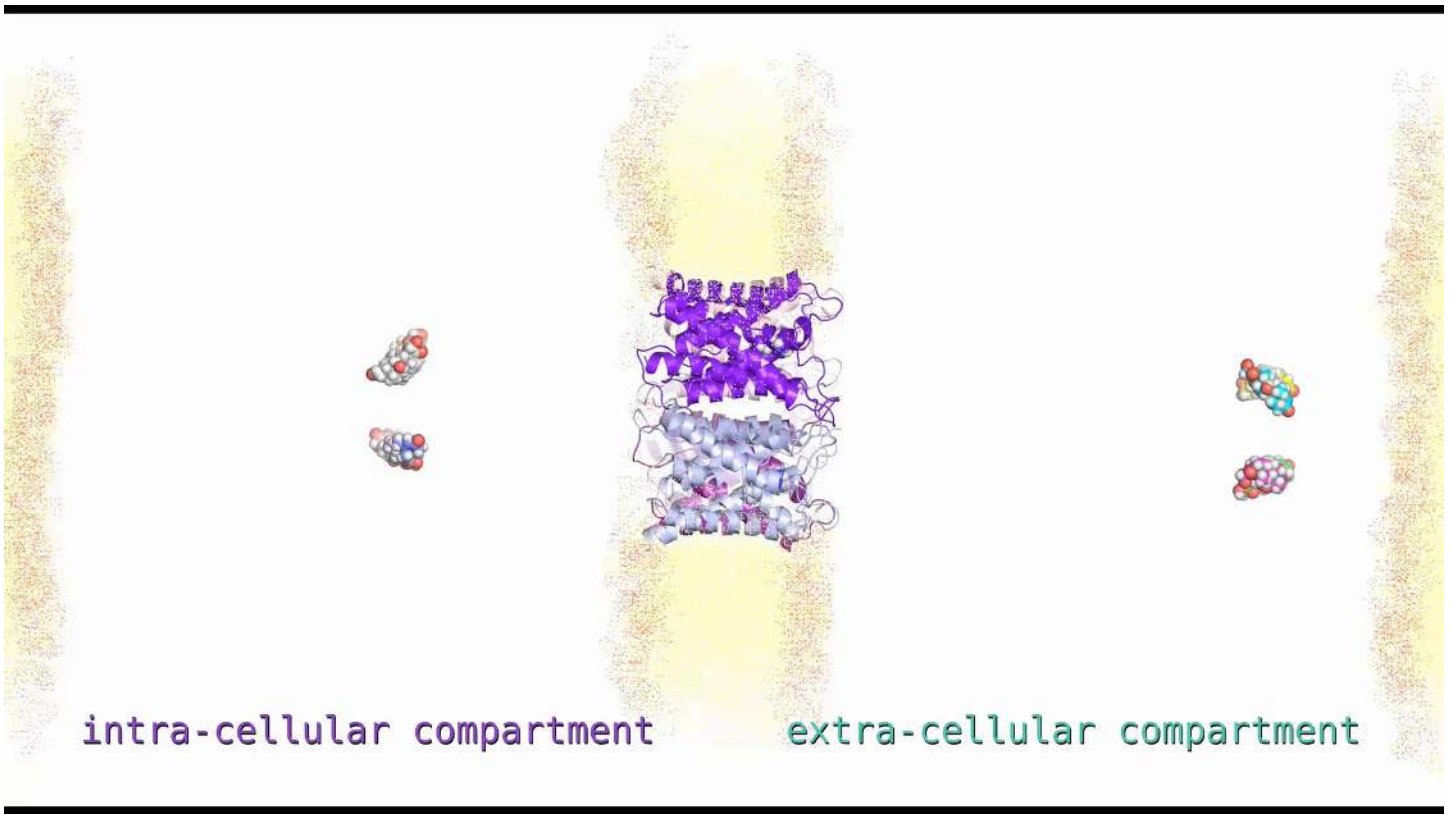
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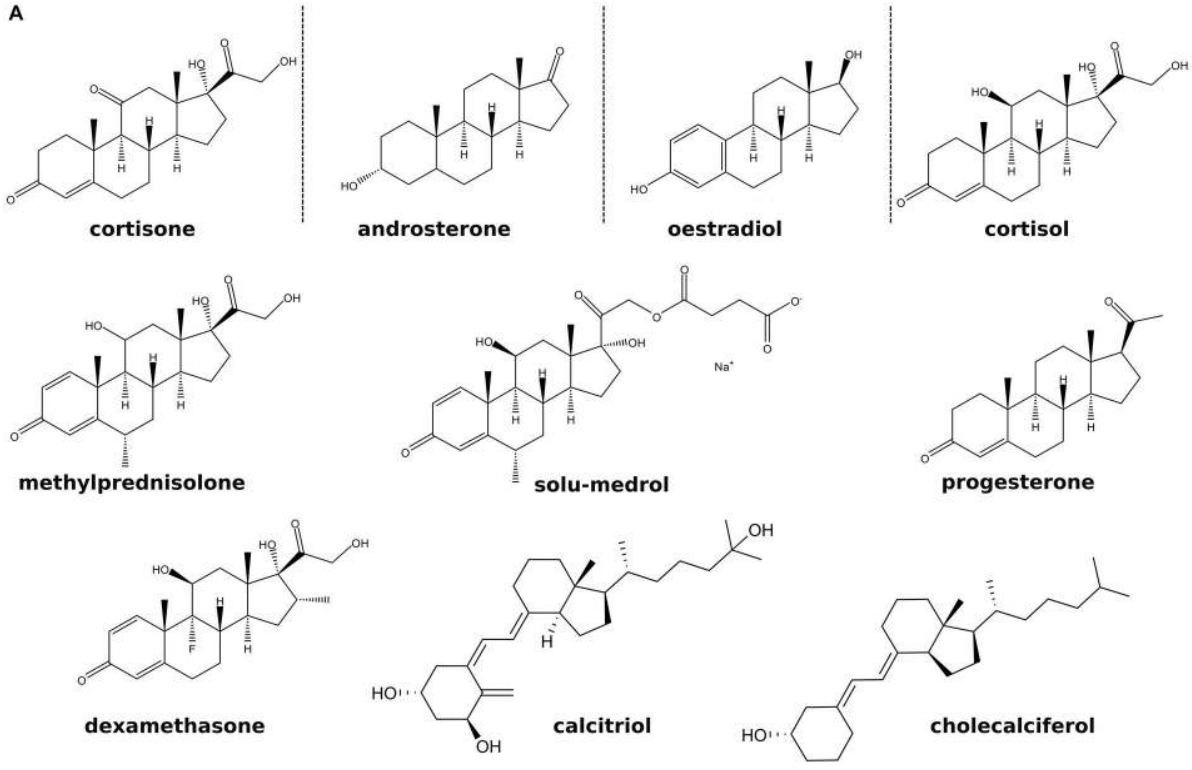
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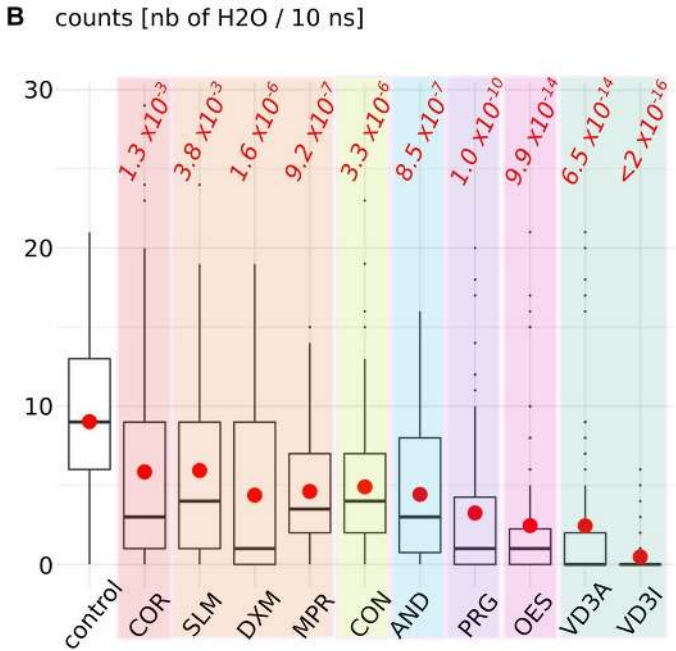
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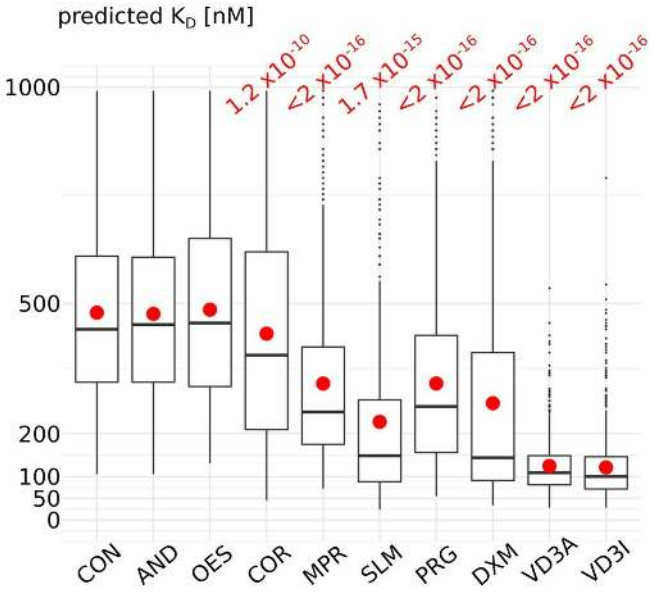
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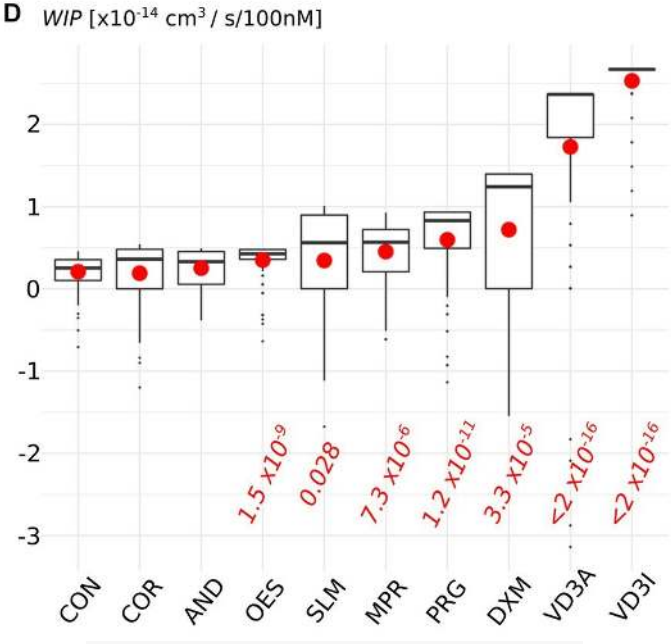


Mom, R.; Réty, S.; Mocquet, V.; Auguin, D. **In Silico Pharmacological Study of AQP2 Inhibition by Steroids Contextualized to Ménière's Disease Treatments.** *Frontiers in Neurology* **2023**, *14*.

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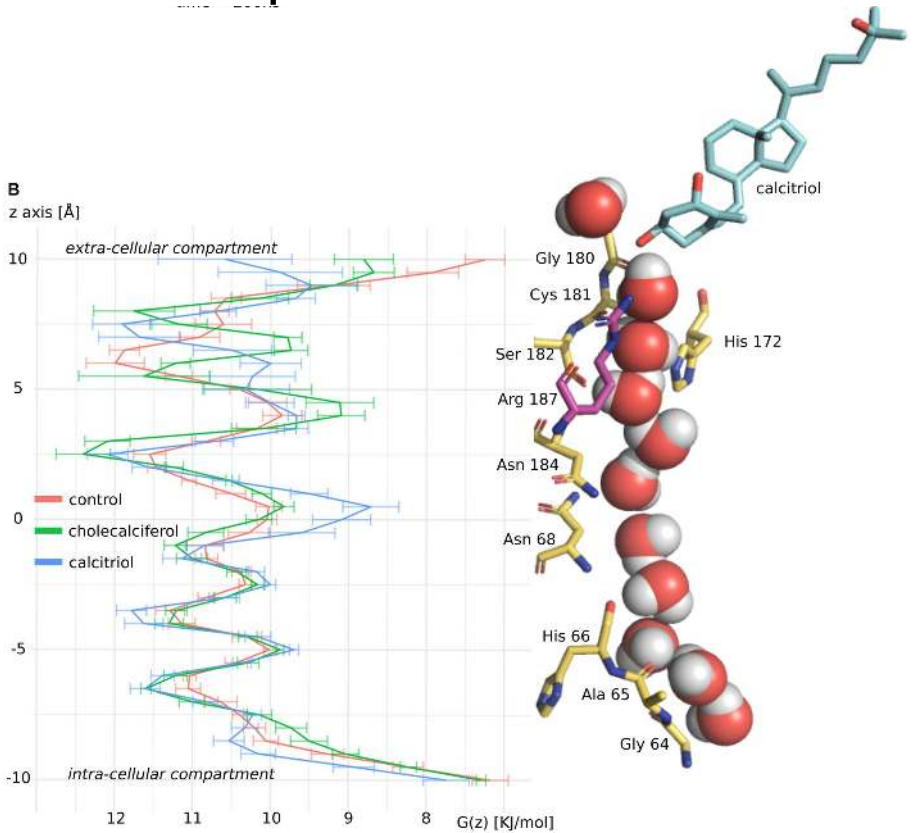


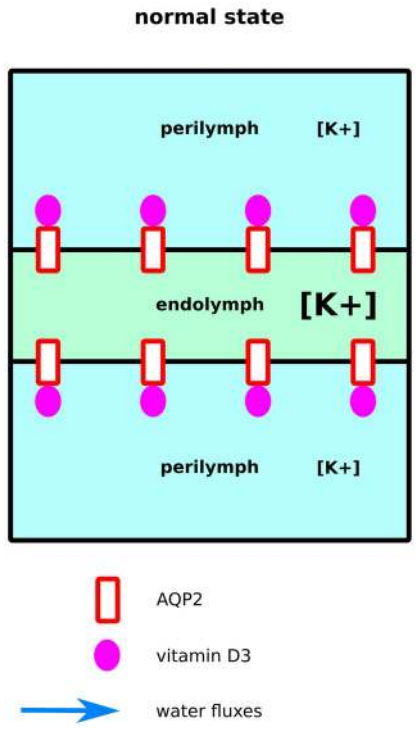


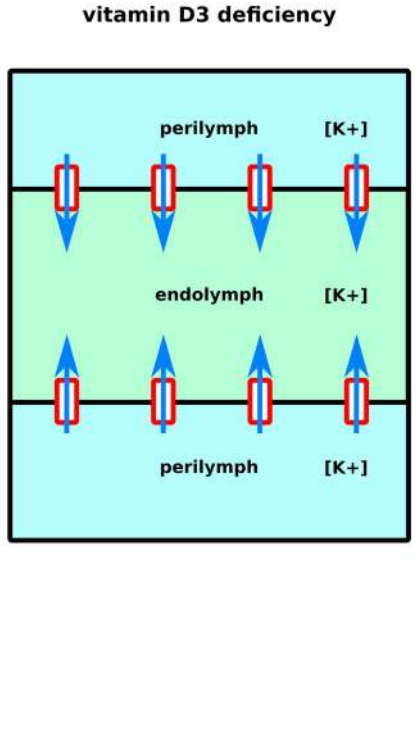
$$\mathrm{WIP} = {}^{(\Delta_{\mathrm{counts}}}/_{\mathrm{K_{Dmedian}})}/_{N_A} * V_m$$

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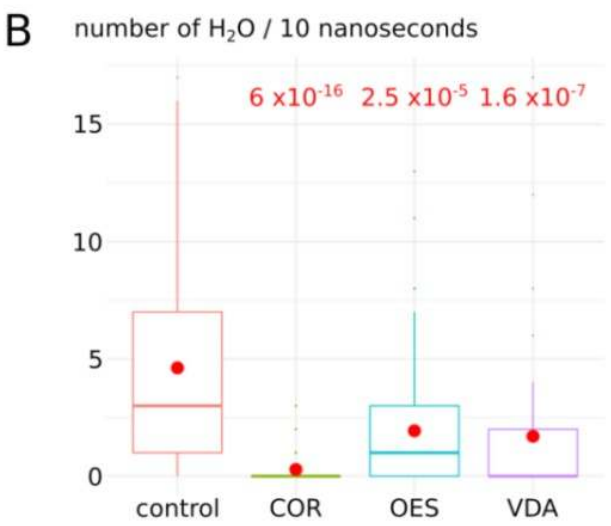


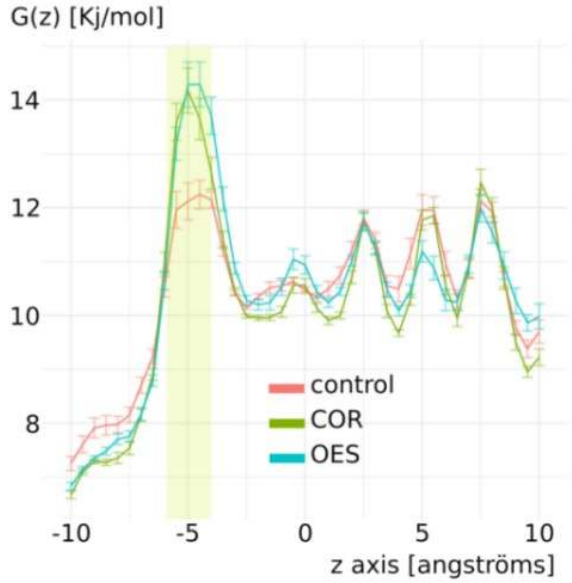




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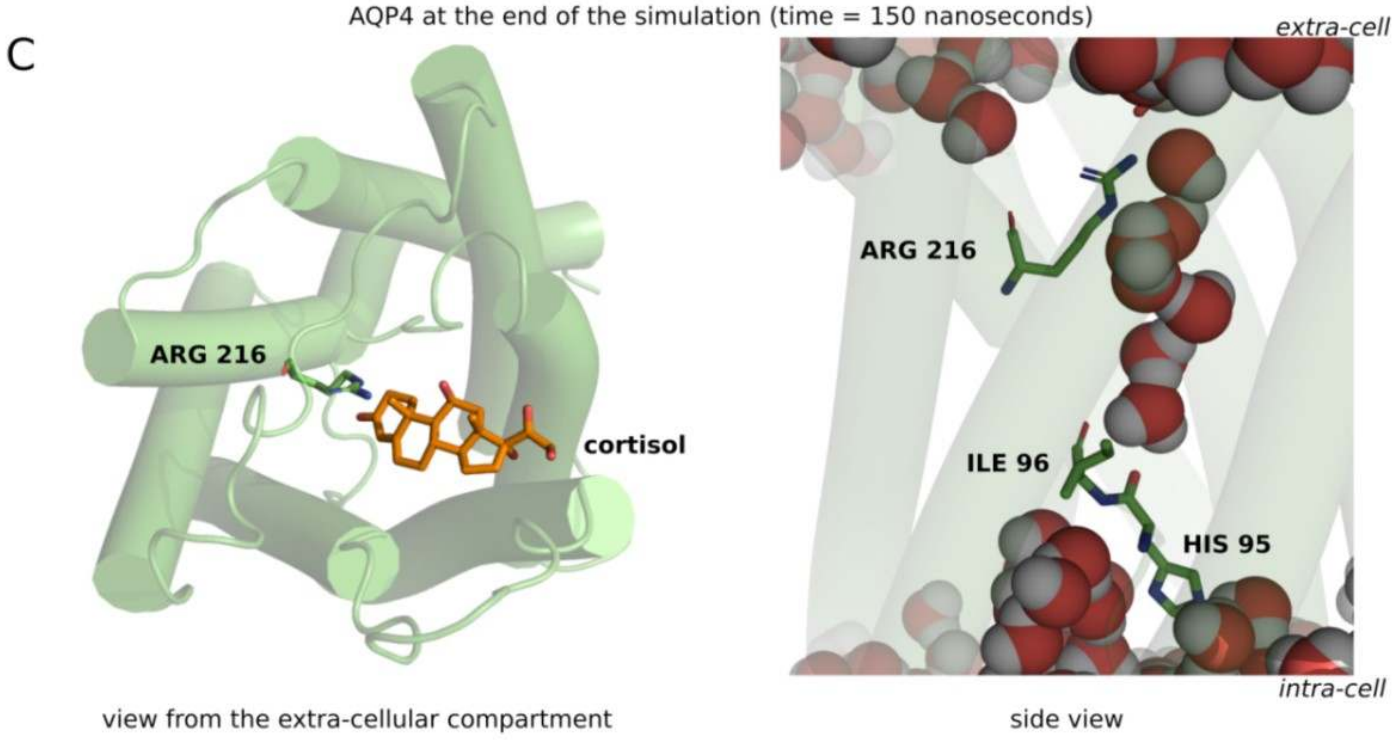
2.2. Modulation by steroids : the role of extra-cellular loops





Preliminary results for human AQP4
(pdb, 3gd8)

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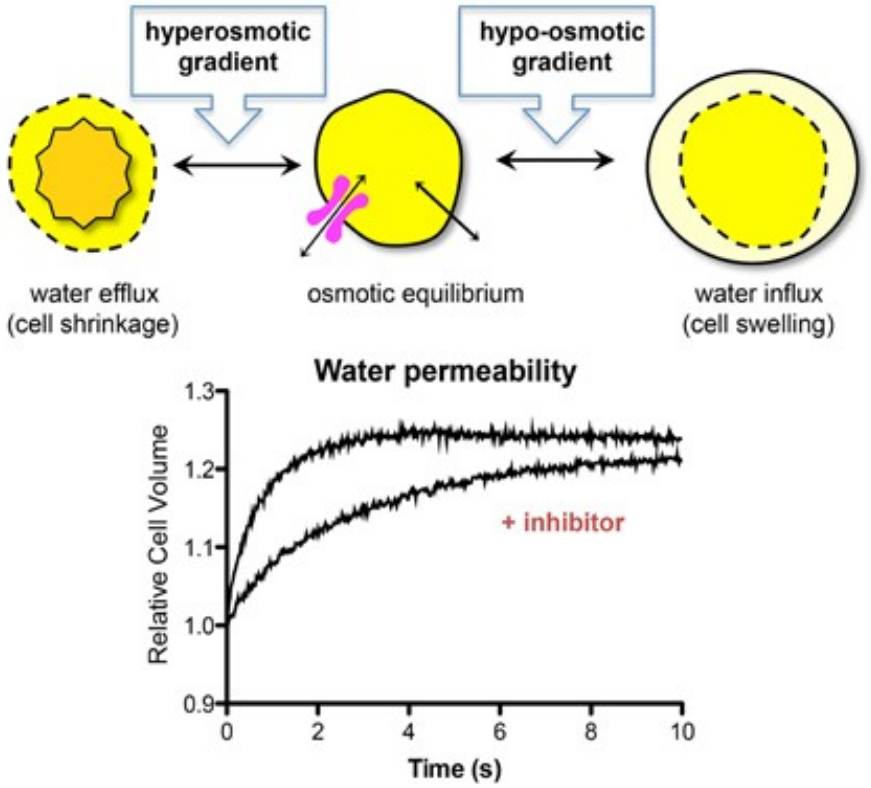


Preliminary results for human AQP4
(pdb, 3gd8)

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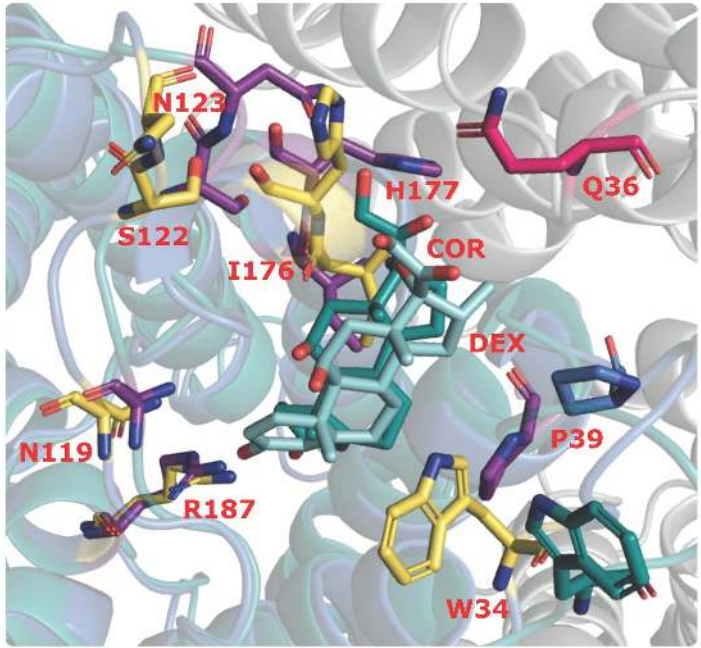


Olga Andrini
Associate Professor at Institut NeuroMyoGène - INMG



Madeira, A.; Moura, T.F.; Soveral, G. **Detecting Aquaporin Function and Regulation.** *Frontiers in Chemistry* **2016**, *4*.

2.2. Modulation by steroids : the role of extra-cellular loops



AQP2 chainD
COR
AQP2 chainB
DEX
Hbonds with COR
Hbonds with DEX
Hbonds with both

AQP0
AQP1
AQP2
AQP3
AQP4
AQP5
AQP6
AQP7
AQP8
AQP9
AQP10
AQP11
AQP12

dexamethasone, methylprednisolone and solu-medrol

synthetic steroids used in treatments

cortisol

glucocorticoids

cortisone

mineralocorticoids

androsterone

androgens

progesterone

prohesterogens

oestradiol

oestrogens

Cholecalciferol and calcitriol

Vitamin D

Sitosterol and 24-epibrassinolide

phytosterols

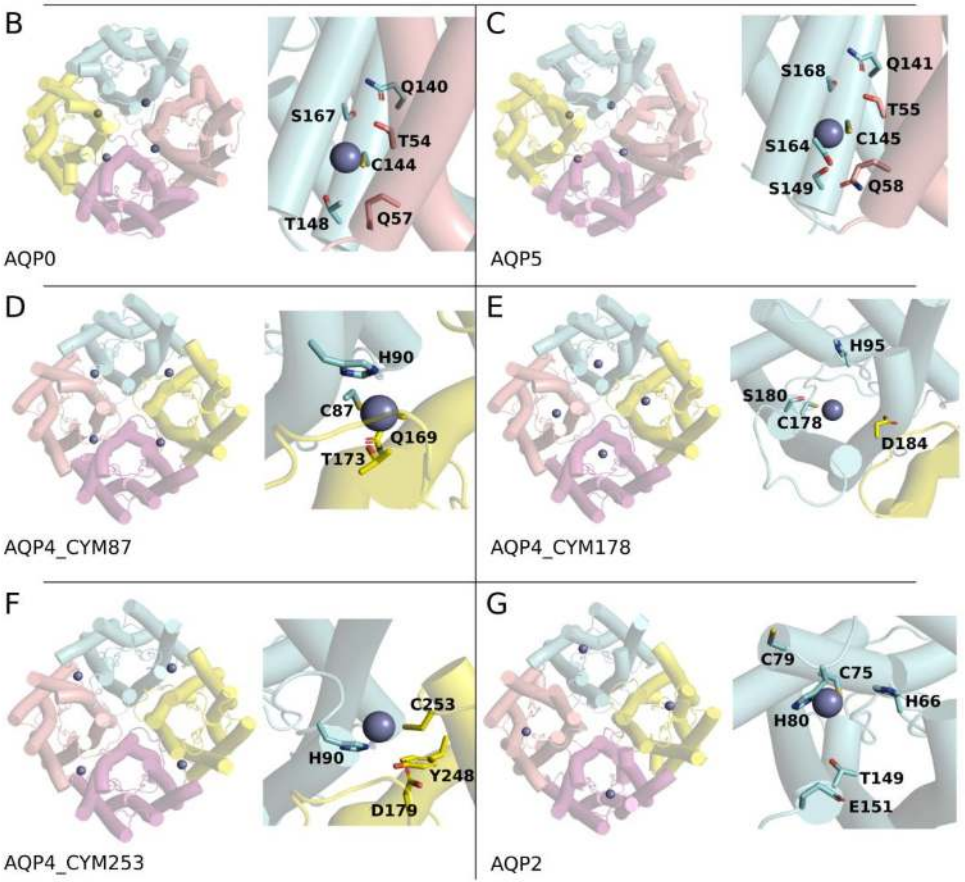
ecdysterone

Steroid hormones from arthropodes

Quercetin, ...

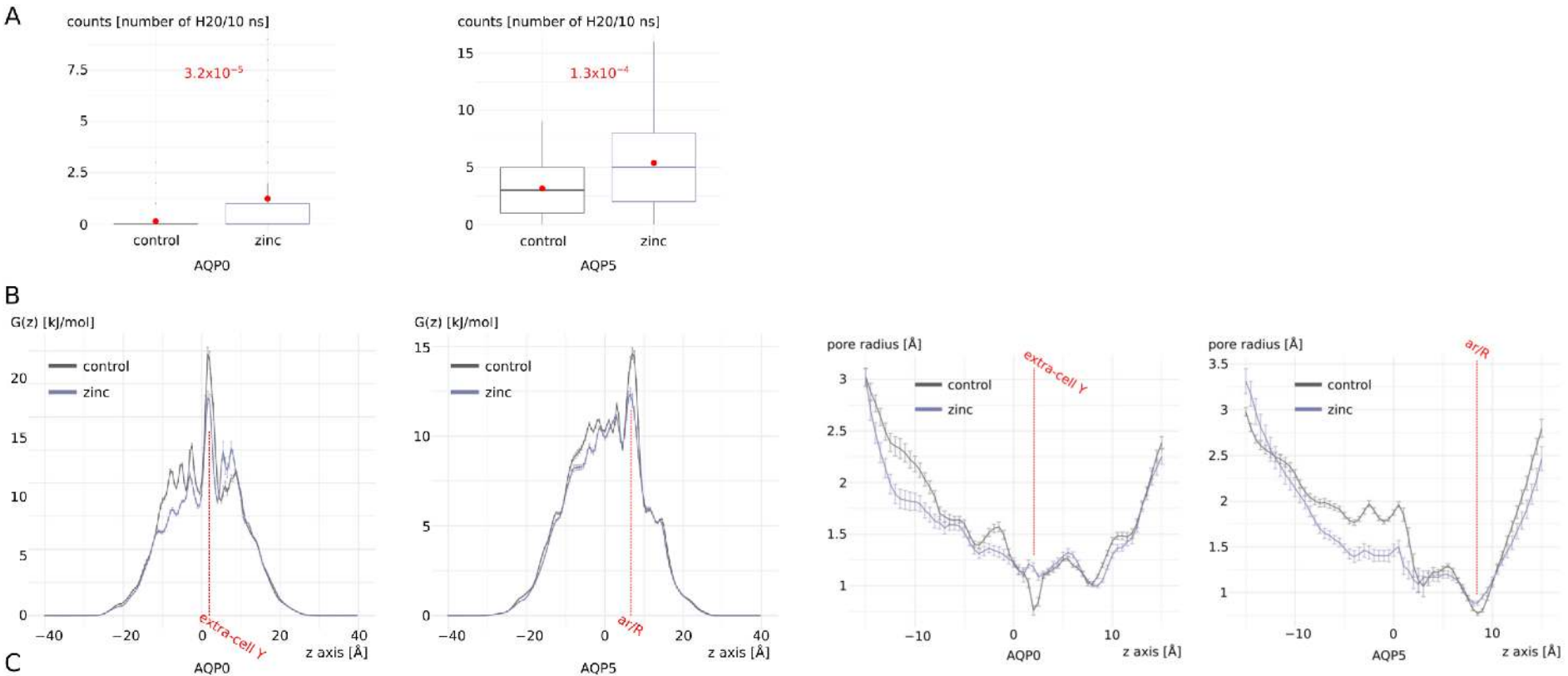
plant secondary metabolites

2.3. Modulation by cations : highlight on cooperativity between monomers



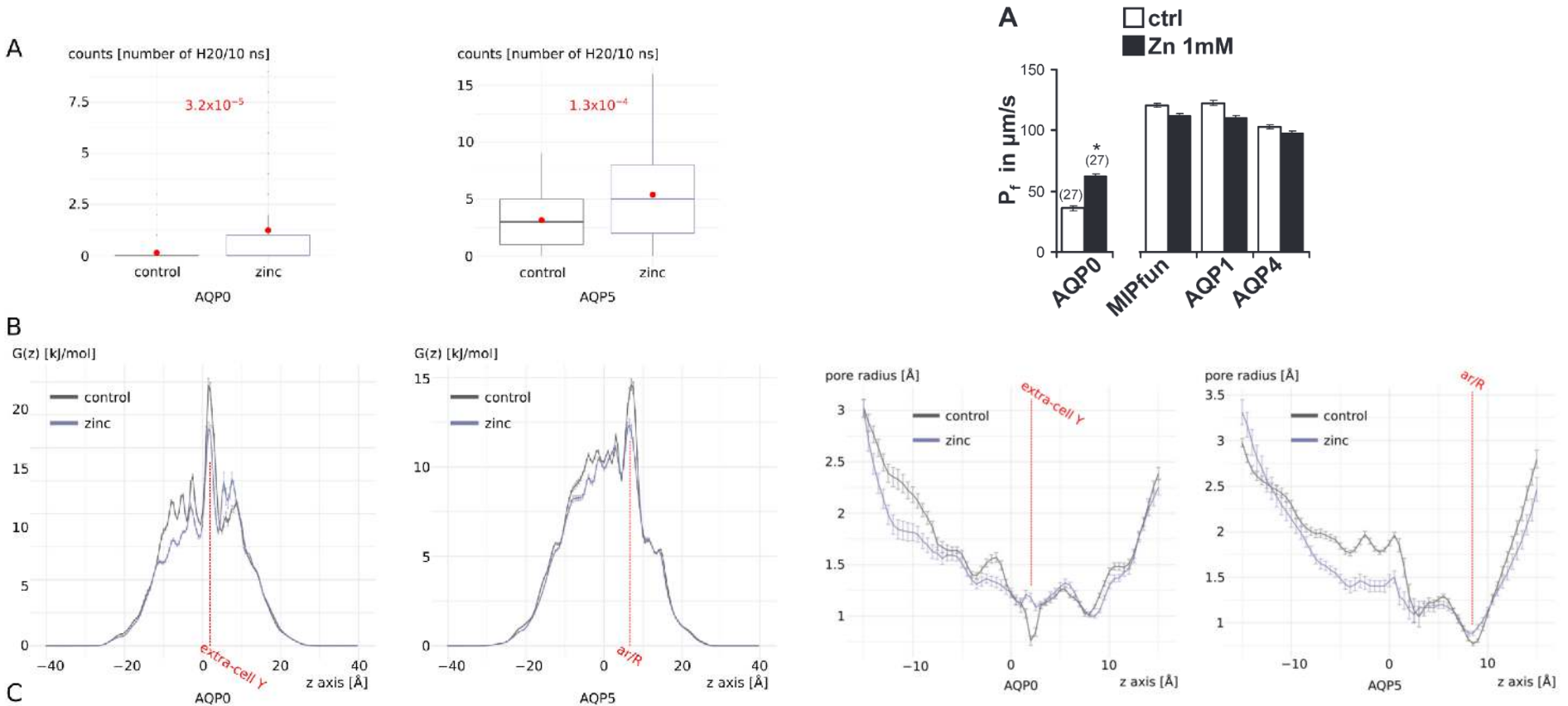
Mom, R.; Réty, S.; Mocquet, V.; Auguin, D. **Deciphering Molecular Mechanisms Involved in the Modulation of Human Aquaporins' Water Permeability by Zinc Cations: A Molecular Dynamics Approach.** *International Journal of Molecular Sciences* **2024**, *25*, 2267, doi:10.3390/ijms25042267.

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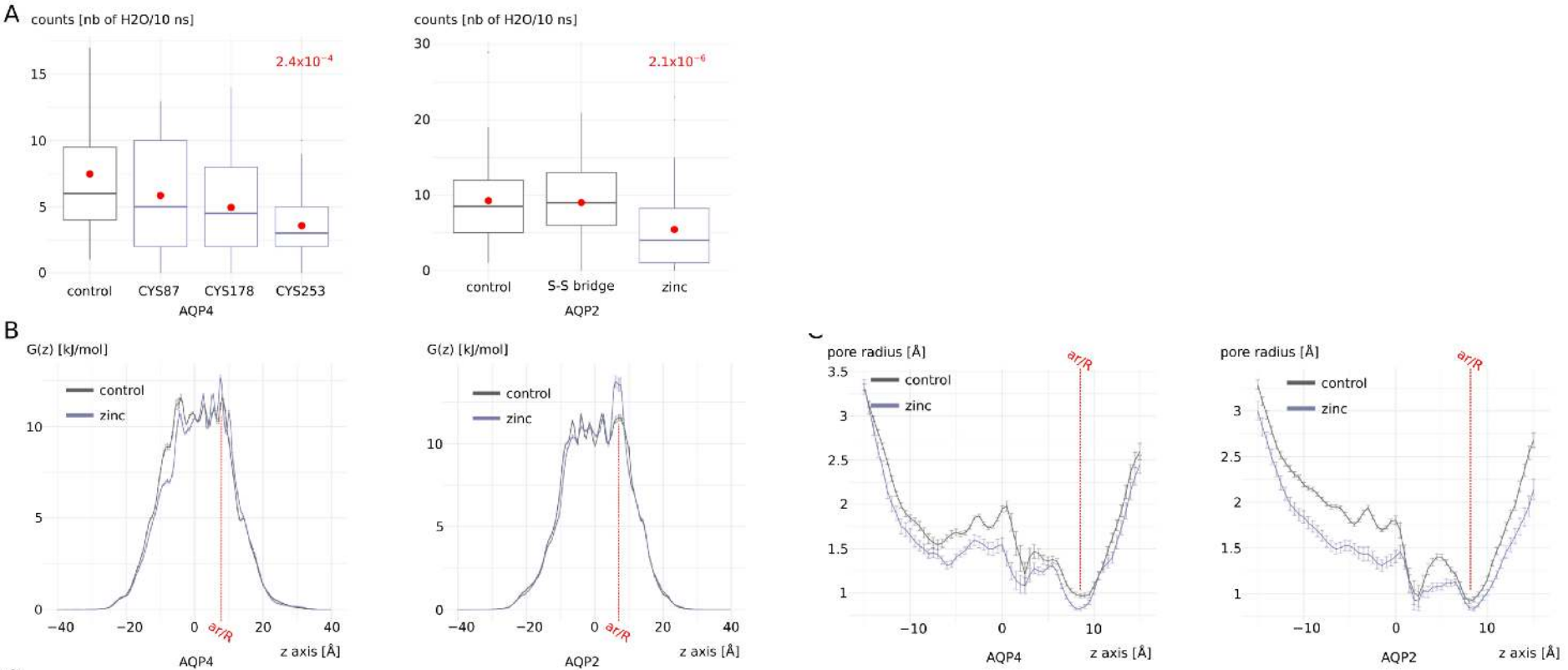
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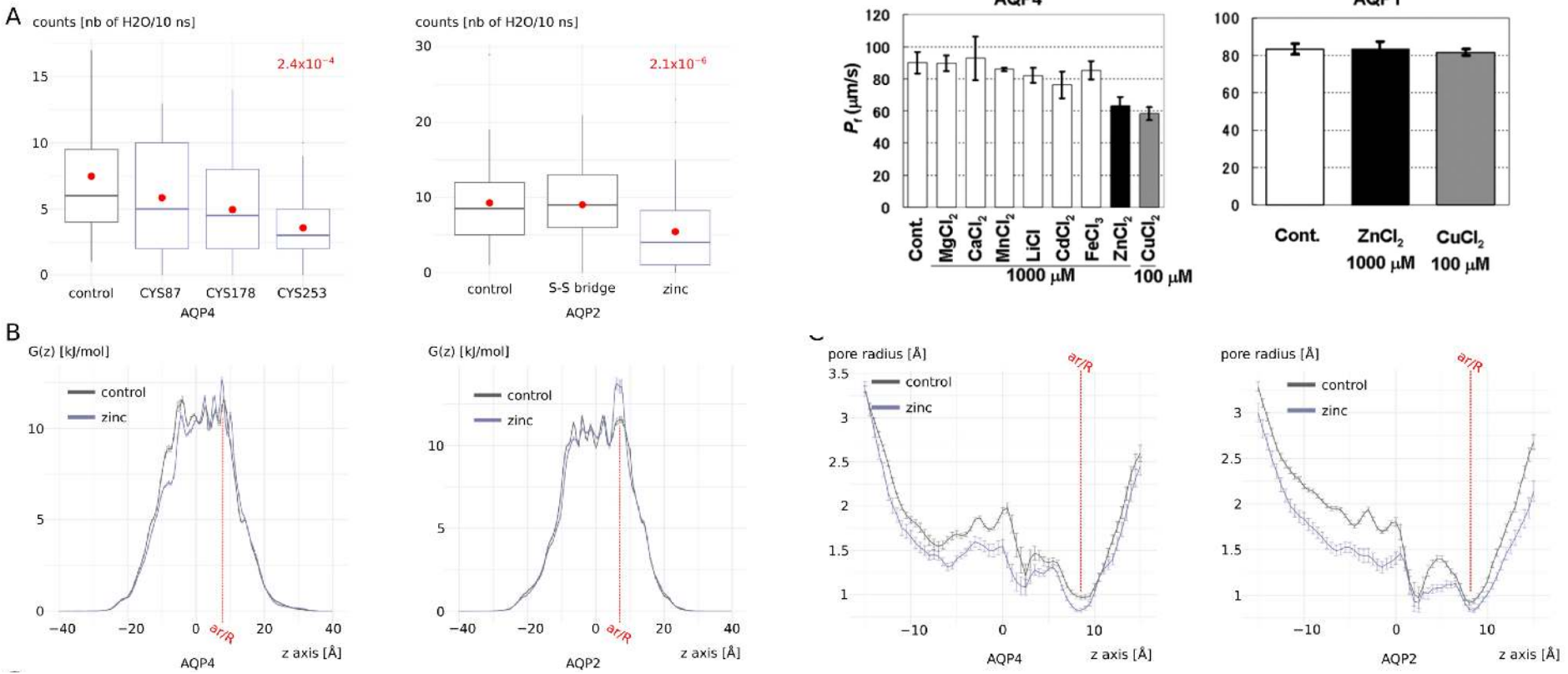
Németh-Cahalan, K.L.; Kalman, K.; Froger, A.; Hall, J.E. **Zinc Modulation of Water Permeability Reveals That Aquaporin 0 Functions as a Cooperative Tetramer.** *Journal of General Physiology* **2007**, *130*, 457–464, doi:10.1085/jgp.200709826.

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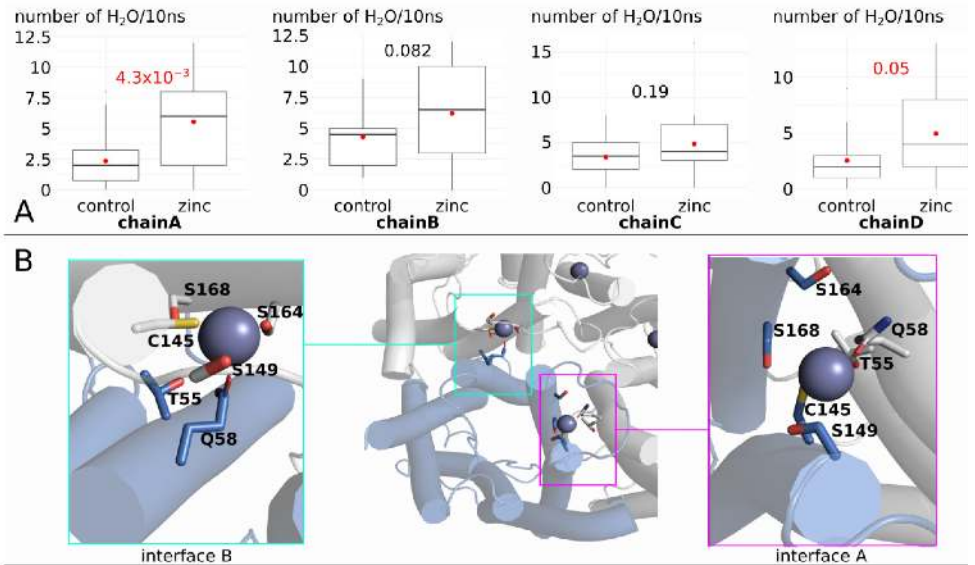
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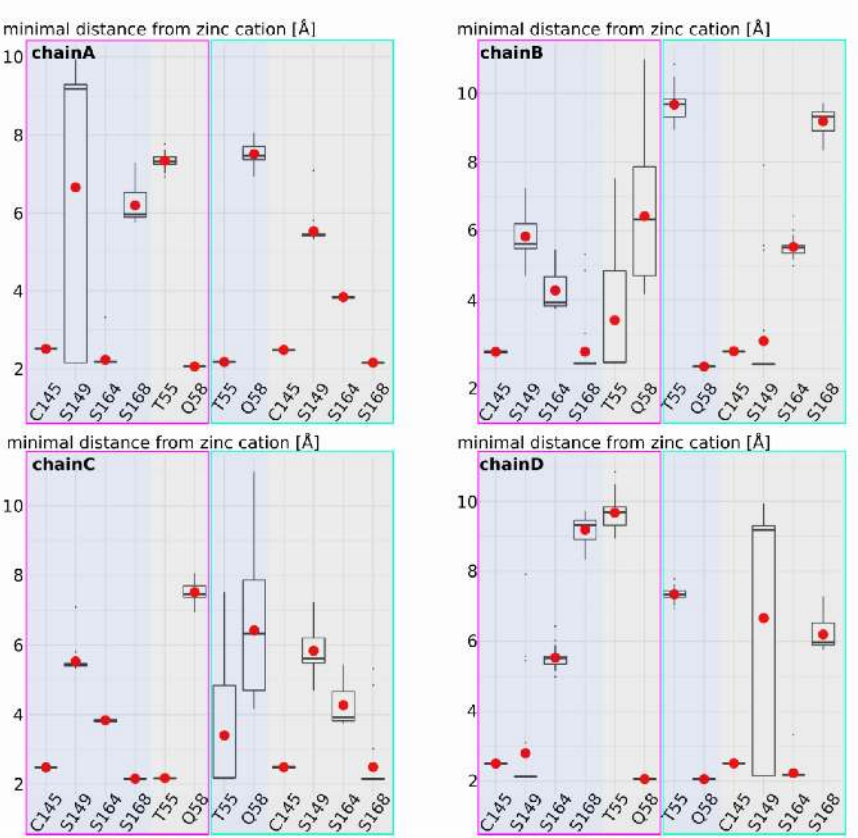


Yukutake, Y.; Hirano, Y.; Suematsu, M.; Yasui, M. **Rapid and Reversible Inhibition of Aquaporin-4 by Zinc.** *Biochemistry* **2009**, *48*, 12059–12061, doi:10.1021/bi901762y.

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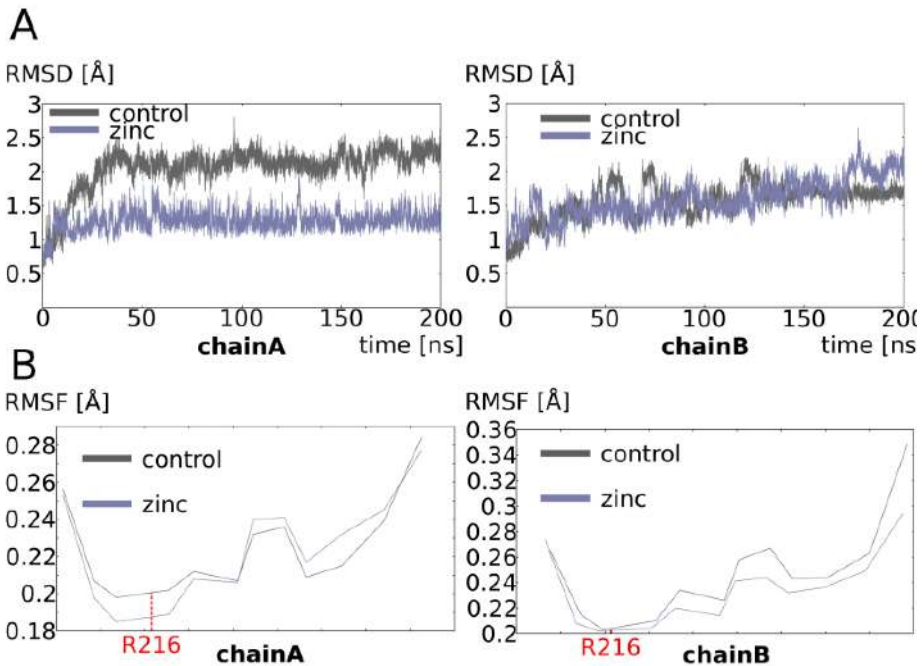


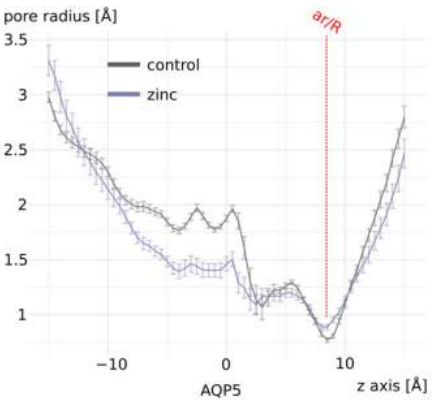
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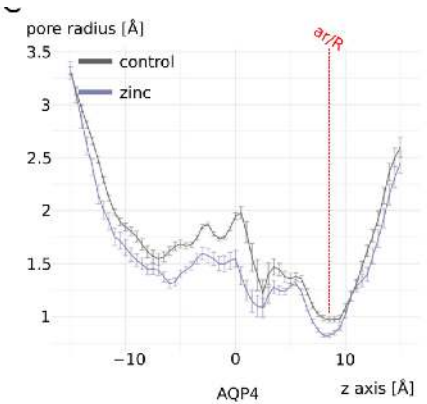


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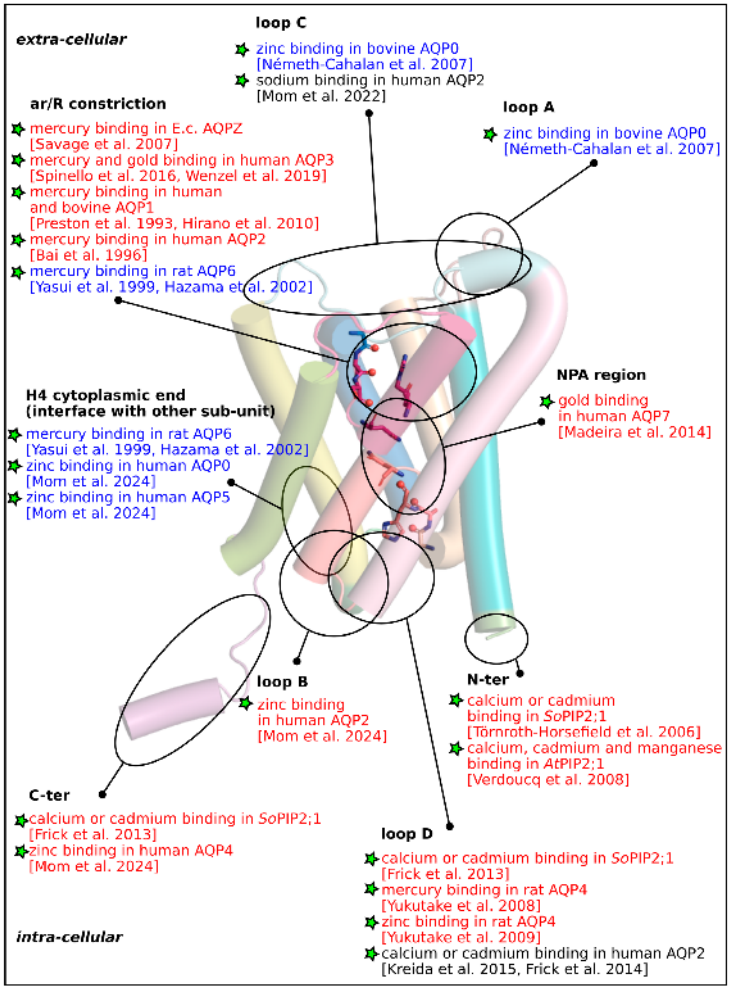






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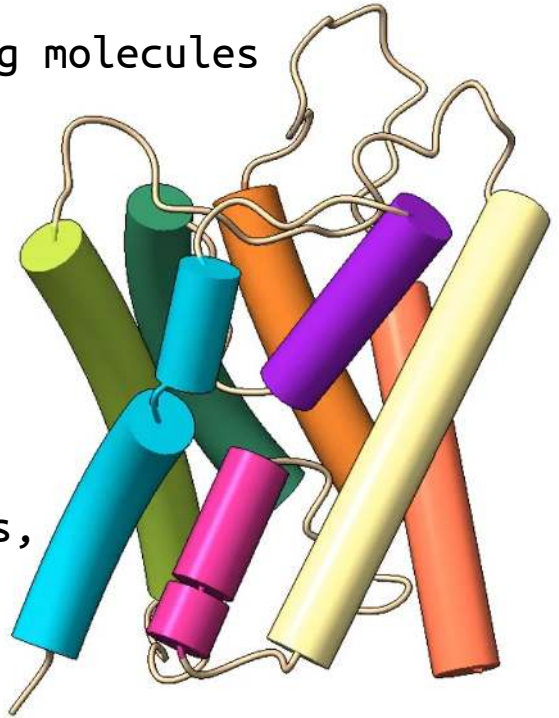
Mom, R.; Mocquet, V.; Auguin, D.; Réty, S. **Aquaporin Modulation by Cations, a Review.** *Current Issues in Molecular Biology* **2024**, *46*, 7955–7975, doi:10.3390/cimb46080470.

3. Conclusions and perspectives

Gene transcription

Extra-cellular loops:
Regulation by small circulating molecules

Post-transcriptional modifications



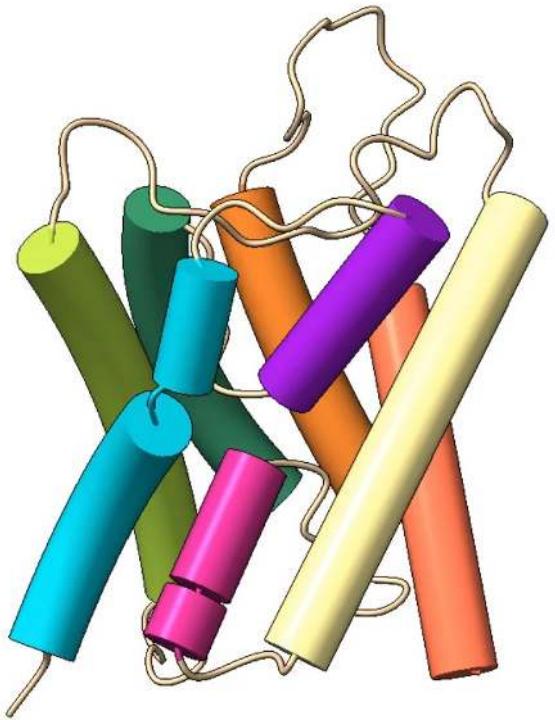
Interface between monomers:
Cooperativity between monomers (regulation by cations, heterotetramerization in plants, Membrane composition

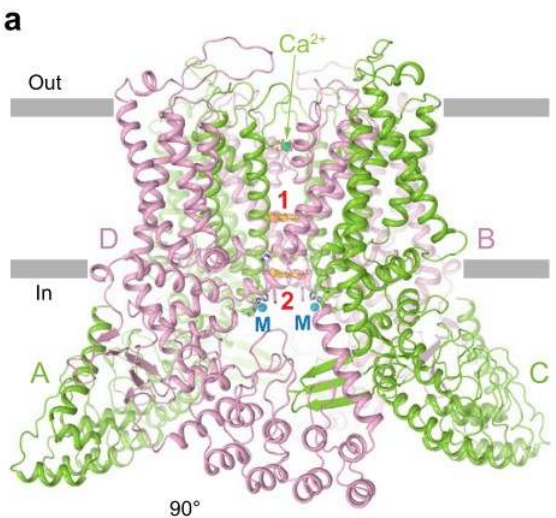
N-ter and C-ter extremities:
Cellular trafficking,
Association with calmodulins

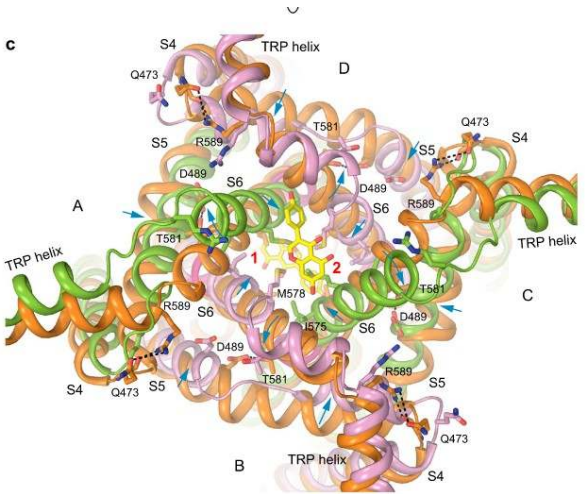
Intra-cellular loop D:
Gating of the channel (calcium and phosphorylation)

3. Conclusions and perspectives

Extra-cellular loops:
Regulation by small circulating molecules

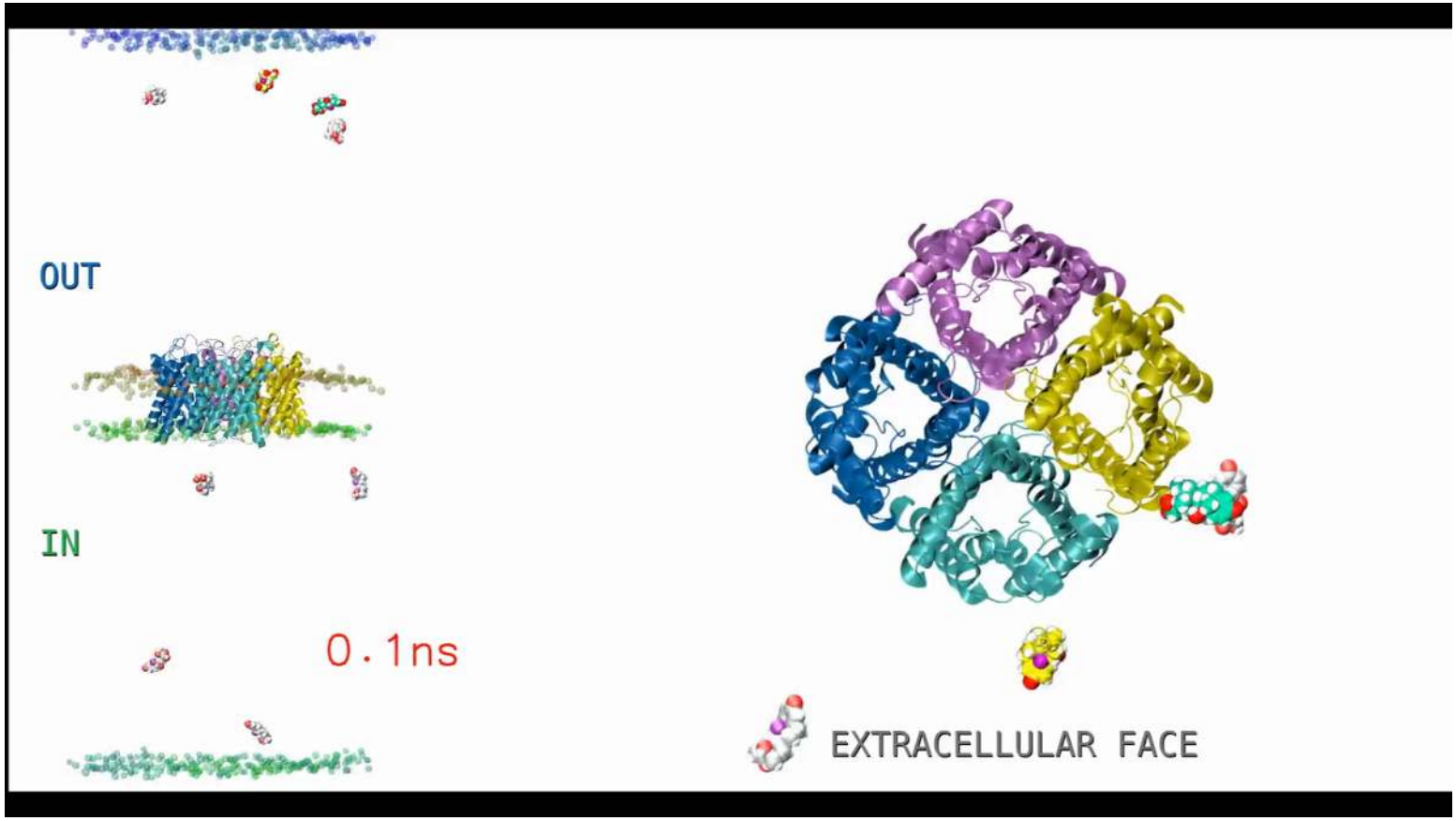






Neuberger, A.; Trofimov, Y.A.; Yelshanskaya, M.V.; Nadezhdin, K.D.; Krylov, N.A.; Efremov, R.G.; Sobolevsky, A.I. **Structural Mechanism of Human Oncochannel TRPV6 Inhibition by the Natural Phytoestrogen Genistein.** *Nat Commun* **2023**, *14*, 2659, doi:10.1038/s41467-023-38352-5.

2.2. Modulation by steroids : the role of extra-cellular loops



Mom, R.; Robert-Paganin, J.; Mom, T.; Chabbert, C.; Réty, S.; Auguin, D. **A Perspective for Ménière's Disease: In Silico Investigations of Dexamethasone as a Direct Modulator of AQP2.** *Biomolecules* **2022**, *12*, 511, doi:10.3390/biom12040511.